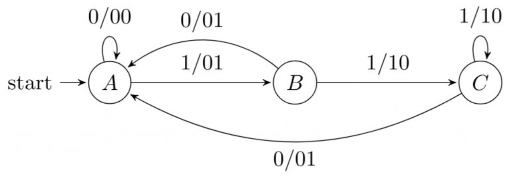


A. outputs the sum of the present and the previous bits of the input
B. outputs 01 whenever the input sequence contains 11
C. outputs 00 whenever the input sequence contains 10
D. none of the above

gatecse-2002 theory-of-computation normal finite-automata

Answer key

6.7.9 Finite Automata: GATE CSE 2002 | Question: 21

We require a four state automaton to recognize the regular expression $(a \mid b)^*abb$

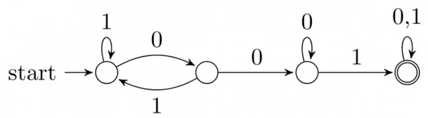
A. Give an NFA for this purpose
B. Give a DFA for this purpose

gatecse-2002 theory-of-computation finite-automata normal descriptive

Answer key

6.7.10 Finite Automata: GATE CSE 2003 | Question: 50

Consider the following deterministic finite state automaton M.



Let S denote the set of seven bit binary strings in which the first, the fourth, and the last bits are1. The number of strings in S that are accepted by M is

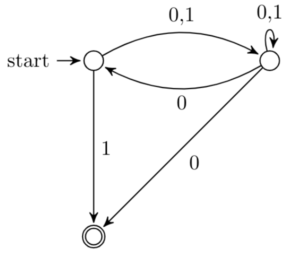
A. 1 B. 5 C. 7 D. 8

gatecse-2003 theory-of-computation finite-automata normal

Answer key

6.7.11 Finite Automata: GATE CSE 2003 | Question: 55

Consider the NFA M shown below.



Let the language accepted by M be L. Let L_1 be the language accepted by the NFA M_1 obtained by changing the accepting state of M to a non-accepting state and by changing the non-accepting states of$M$ to accepting states. Which of the following statements is true?

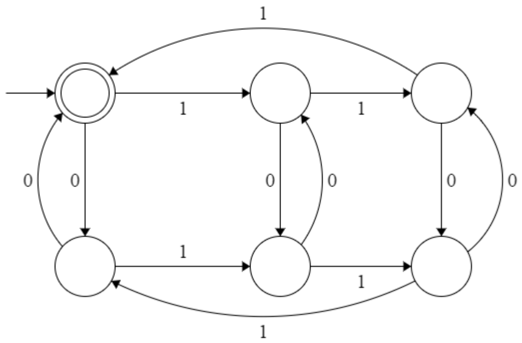
A. $L_1 = \{0,1\}^* - L$ B. $L_1 = \{0,1\}^*$ C. $L_1 \subseteq L$ D. $L_1 = L$

gatecse-2003 theory-of-computation finite-automata normal

Answer key

6.7.12 Finite Automata: GATE CSE 2004 | Question: 86

The following finite state machine accepts all those binary strings in which the number of 1's and 0's are respectively:



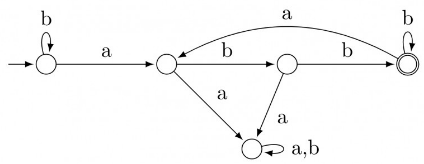
A. divisible by 3 and 2
B. odd and even
C. even and odd
D. divisible by 2 and 3

gatecse-2004 theory-of-computation finite-automata easy

Answer key

6.7.13 Finite Automata: GATE CSE 2005 | Question: 53

Consider the machine M:



The language recognized by M is:

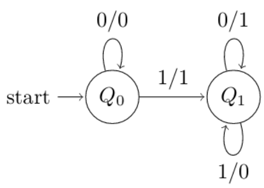
A. $\{w \in \{a,b\}^* \mid \text{every a in } w \text{ is followed by exactly two } b\text{'s}\}$
B. $\{w \in \{a,b\}^* \mid \text{every a in } w \text{ is followed by at least two } b\text{'s}\}$
C. $\{w \in \{a,b\}^* \mid w \text{ contains the substring '}abb\text{'}\}$
D. $\{w \in \{a,b\}^* \mid w \text{ does not contain '}aa\text{' as a substring}\}$

gatecse-2005 theory-of-computation finite-automata normal

Answer key

6.7.14 Finite Automata: GATE CSE 2005 | Question: 63

The following diagram represents a finite state machine which takes as input a binary number from the least significant bit.



Which of the following is TRUE?

A. It computes 1's complement of the input number

B. It computes 2's complement of the input number

C. It increments the input number

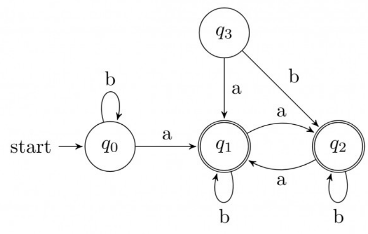
D. it decrements the input number

gatecse-2005 theory-of-computation finite-automata easy

Answer key☟

6.7.15 Finite Automata: GATE CSE 2007 | Question: 74

Consider the following Finite State Automaton:



The language accepted by this automaton is given by the regular expression

A. $b^*ab^*ab^*ab^*$ B. $(a+b)^*$ C. $b^*a(a+b)^*$ D. $b^*ab^*ab^*$

gatecse-2007 theory-of-computation finite-automata normal

Answer key☟

6.7.16 Finite Automata: GATE CSE 2008 | Question: 49

Given below are two finite state automata ($\rightarrow$ indicates the start state and F indicates a final state)

Y

	a	b
$\rightarrow 1$	1	2
$2(F)$	2	1

Z

	a	b
$\rightarrow 1$	2	2
$2(F)$	1	1

Which of the following represents the product automaton $Z \times Y$?

A.

	a	b
$\rightarrow P$	S	R
Q	R	S
R(F)	Q	P
S	Q	P

B.

	a	b
$\rightarrow P$	S	Q
Q	R	S
R(F)	Q	P
S	P	Q

C.

	a	b
$\rightarrow P$	Q	S
Q	R	S
R(F)	Q	P
S	Q	P

D.

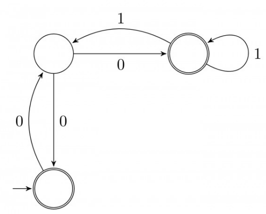
	a	b
$\rightarrow P$	S	Q
Q	S	R
R(F)	Q	P
S	Q	P

gatecse-2008 normal theory-of-computation finite-automata

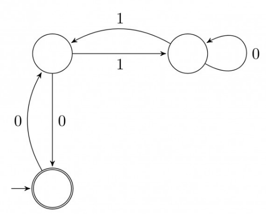
Answer key☟

6.7.17 Finite Automata: GATE CSE 2008 | Question: 52

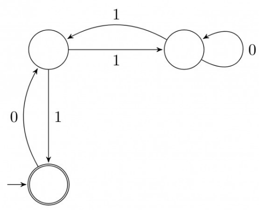
Match the following NFAs with the regular expressions they correspond to:



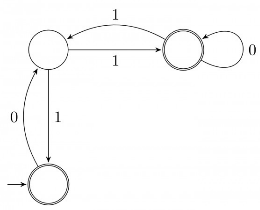
P



Q



R



S

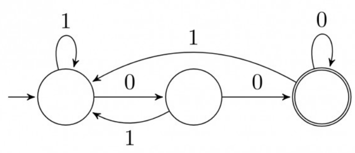
1. $\epsilon + 0(01^*1 + 00)^*01^*$
2. $\epsilon + 0(10^*1 + 00)^*0$
3. $\epsilon + 0(10^*1 + 10)^*1$
4. $\epsilon + 0(10^*1 + 10)^*10^*$

A. $P-2, Q-1, R-3, S-4$
B. $P-1, Q-3, R-2, S-4$
C. $P-1, Q-2, R-3, S-4$
D. $P-3, Q-2, R-1, S-4$

gatecse-2008 theory-of-computation finite-automata normal

Answer key☟

6.7.18 Finite Automata: GATE CSE 2009 | Question: 41



The above DFA accepts the set of all strings over $\{0, 1\}$ that

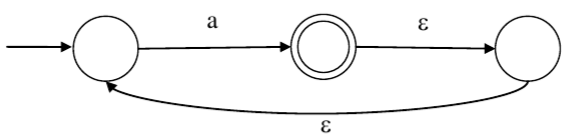
A. begin either with 0 or 1.
B. end with 0.
C. end with 00.
D. contain the substring 00.

gatecse-2009 theory-of-computation finite-automata easy

Answer key☟

6.7.19 Finite Automata: GATE CSE 2012 | Question: 12

What is the complement of the language accepted by the NFA shown below?
Assume $\Sigma = \{a\}$ and ϵ is the empty string.



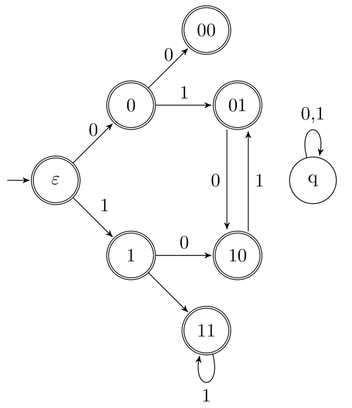
A. ϕ
B. $\{\epsilon\}$
C. a^*
D. $\{a, \epsilon\}$

gatecse-2012 finite-automata easy theory-of-computation

Answer key☟

6.7.20 Finite Automata: GATE CSE 2012 | Question: 46

Consider the set of strings on $\{0, 1\}$ in which, *every substring of* 3 *symbols* has at most *two* zeros. For example, 001110 and 011001 are in the language, but 100010 is not. All strings of length less than 3 are also in the language. A partially completed DFA that accepts this language is shown below.



The missing arcs in the DFA are:

A.

	00	01	10	11	q
00	1	0			
01				1	
10	0				
11			0		

B.

	00	01	10	11	q
00		0			1
01		1			
10				0	
11		0			

C.

	00	01	10	11	q
00		1			0
01		1			
10			0		
11		0			

D.

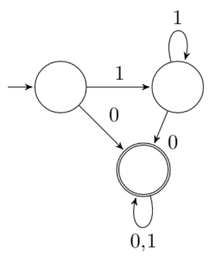
	00	01	10	11	q
00		1			0
01				1	
10	0				
11			0		

gatecse-2012 theory-of-computation finite-automata normal

Answer key☟

6.7.21 Finite Automata: GATE CSE 2013 | Question: 33

Consider the DFA A given below.



Which of the following are **FALSE**?

1. Complement of $L(A)$ is context-free.
2. $L(A) = L((11^*0+0)(0+1)^*0^*1^*)$
3. For the language accepted by A, A is the minimal DFA.
4. A accepts all strings over $\{0,1\}$ of length at least 2.

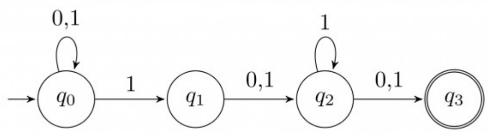
A. 1 and 3 only B. 2 and 4 only C. 2 and 3 only D. 3 and 4 only

gatecse-2013 theory-of-computation finite-automata normal

Answer key☟

6.7.22 Finite Automata: GATE CSE 2014 | Set 1 | Question: 16

Consider the finite automaton in the following figure:



What is the set of reachable states for the input string 0011?

A. $\{q_0, q_1, q_2\}$
B. $\{q_0, q_1\}$
C. $\{q_0, q_1, q_2, q_3\}$
D. $\{q_3\}$

gatecse-2014-set1 theory-of-computation finite-automata easy

Answer key☟

6.7.23 Finite Automata: GATE CSE 2016 | Set 2 | Question: 42

Consider the following two statements:

I. If all states of an **NFA** are accepting states then the language accepted by the **NFA** is Σ^*.
II. There exists a regular language A such that for all languages B, $A \cap B$ is regular.

Which one of the following is **CORRECT**?

A. Only **I** is true
B. Only **II** is true
C. Both **I** and **II** are true
D. Both **I** and **II** are false

gatecse-2016-set2 theory-of-computation finite-automata normal

Answer key☟

6.7.24 Finite Automata: GATE CSE 2017 | Set 2 | Question: 39

Let δ denote the transition function and $\hat{\delta}$ denote the extended transition function of the ϵ-NFA whose transition table is given below:

δ	ϵ	a	b
$\$ \rightarrow \$q_0$	$\{q_2\}$	$\{q_1\}$	$\{q_0\}$
q_1	$\{q_2\}$	$\{q_2\}$	$\{q_3\}$
q_2	$\{q_0\}$	$\emptyset$	$\emptyset$
q_3	$\emptyset$	$\emptyset$	$\{q_2\}$

Then $\hat{\delta}(q_2, aba)$ is

A. $\emptyset$
B. $\{q_0, q_1, q_3\}$
C. $\{q_0, q_1, q_2\}$
D. $\{q_0, q_2, q_3\}$

gatecse-2017-set2 theory-of-computation finite-automata

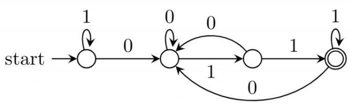
Answer key☟

6.7.25 Finite Automata: GATE CSE 2021 | Set 1 | Question: 38

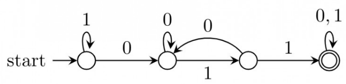
Consider the following language:

$$L = \{w \in \{0,1\}^* \mid w \text{ ends with the substring } 011\}$$

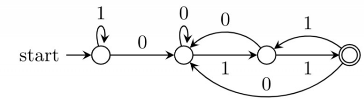
Which one of the following deterministic finite automata accepts L?



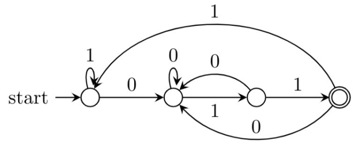
A.



B.



C.



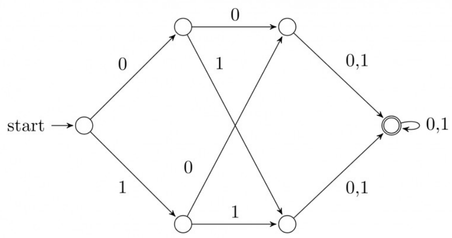
D.

gatecse-2021-set1 theory-of-computation finite-automata two-marks

Answer key

6.7.26 Finite Automata: GATE CSE 2021 | Set 2 | Question: 17

Consider the following deterministic finite automaton (DFA)



The number of strings of length 8 accepted by the above automaton is ___________

gatecse-2021-set2 numerical-answers theory-of-computation finite-automata one-mark

Answer key

6.7.27 Finite Automata: GATE CSE 2021 | Set 2 | Question: 28

Suppose we want to design a synchronous circuit that processes a string of 0's and 1's. Given a string, it produces another string by replacing the first 1 in any subsequence of consecutive 1's by a 0. Consider the following example.

$$\begin{array}{ll} \text{Input sequence:} & 00100011000011100 \\ \text{Output sequence:} & 00000001000001100 \end{array}$$

A *Mealy Machine* is a state machine where both the next state and the output are functions of the present state and the current input.

The above mentioned circuit can be designed as a two-state Mealy machine. The states in the Mealy machine can be represented using Boolean values 0 and 1. We denote the current state, the next state, the next incoming bit, and the output bit of the Mealy machine by the variables s, t, b and y respectively.

Assume the initial state of the Mealy machine is 0.

What are the Boolean expressions corresponding to t and y in terms of s and b?

A. $t = s + b$
 $y = sb$

B. $t = b$
 $y = sb$

C. $t = b$
 $y = s\bar{b}$

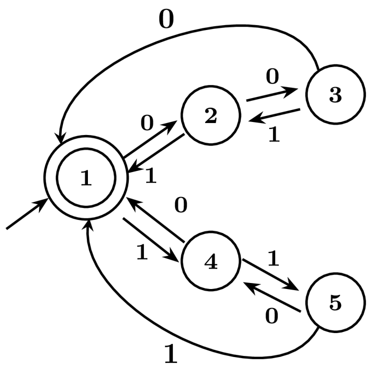
D. $t = s + b$
 $y = s\bar{b}$

gatecse-2021-set2 theory-of-computation finite-automata two-marks

Answer key

6.7.28 Finite Automata: GATE CSE 2024 | Set 1 | Question: 40

Consider the 5 -state DFA. M accepting the language $L(M) \subset (0+1)^*$ shown below. For any string $w \in (0+1)^*$ let $n_0(w)$ be the number of $0's$ in w and $n_1(w)$ be the number of 1 's in w.



Which of the following statements is/are FALSE?

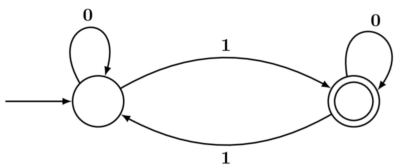
A. States 2 and 4 are distinguishable in M
B. States 3 and 4 are distinguishable in M
C. States 2 and 5 are distinguishable in M
D. Any string w with $n_0(w) = n_1(w)$ is in $L(M)$

gatecse-2024-set1 multiple-selects theory-of-computation finite-automata two-marks

Answer key

6.7.29 Finite Automata: GATE CSE 2024 | Set 2 | Question: 12

Which one of the following regular expressions is equivalent to the language accepted by the DFA given below?



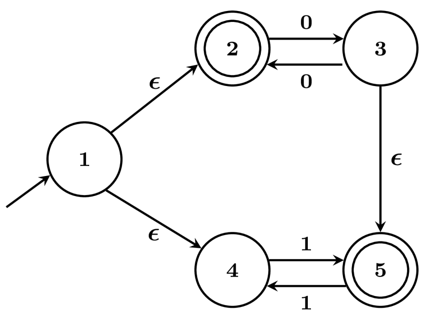
A. $0^*1(0+10^*1)^*$
B. $0^*(10^*11)^*0^*$
C. $0^*1(010^*1)^*0^*$
D. $0(1+0^*10^*1)^*0^*$

gatecse-2024-set2 theory-of-computation finite-automata one-mark

Answer key

6.7.30 Finite Automata: GATE CSE 2024 | Set 2 | Question: 31

Let M be the 5-state NFA with ϵ-transitions shown in the diagram below.



Which one of the following regular expressions represents the language accepted by M?

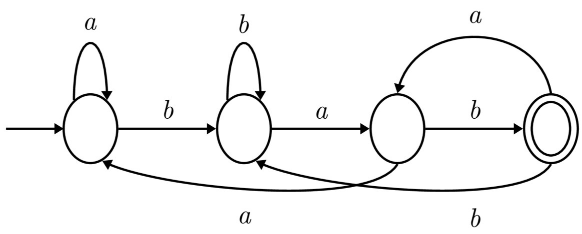
A. $(00)^* + 1(11)^*$
B. $0^* + (1+0(00)^*)(11)^*$
C. $(00)^* + (1+(00)^*)(11)^*$
D. $0^+ + 1(11)^* + 0(11)^*$

gatecse-2024-set2 theory-of-computation finite-automata two-marks

Answer key

6.7.31 Finite Automata: GATE CSE 2025 | Set 1 | Question: 40

Consider the following deterministic finite automaton (DFA) defined over the alphabet, $\Sigma = \{a, b\}$. Identify which of the following language(s) is/are accepted by the given DFA.



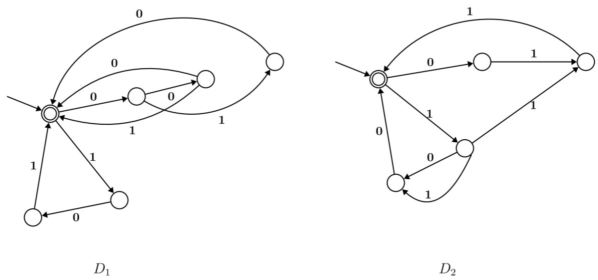
A. The set of all strings containing an even number of b 's.
B. The set of all strings containing the pattern bab.
C. The set of all strings ending with the pattern bab.
D. The set of all strings not containing the pattern aba.

gatecse2025-set1 theory-of-computation finite-automata multiple-selects easy two-marks

Answer key

6.7.32 Finite Automata: GATE CSE 2026 | Set 2 | Question: 37

Consider the following two finite automata D_1 and D_2.



D_1 D_2

Which of the following statements is/are true?

A. $L\left(D_1\right) = L\left(D_2\right)$
B. $L\left(D_1\right)$ is a proper subset of $L\left(D_2\right)$
C. $L\left(D_1\right) \cap L\left(D_2\right) = \{\epsilon\}$
D. $\left(L\left(D_1\right) \cup L\left(D_2\right)\right)^*$ consists of all strings in $\{0, 1\}^*$ whose length is divisible by 3

gatecse-2026-set2 theory-of-computation finite-automata multiple-selects two-marks

Answer key

6.7.33 Finite Automata: GATE IT 2004 | Question: 41

Let $M = (K, \Sigma, \sigma, s, F)$ be a finite state automaton, where

$K = \{A, B\}, \Sigma = \{a, b\}, s = A, F = \{B\}$,
$\sigma(A, a) = A, \sigma(A, b) = B, \sigma(B, a) = B$ and $\sigma(B, b) = A$

A grammar to generate the language accepted by M can be specified as $G = (V, \Sigma, R, S)$, where $V = K \cup \Sigma$, and $S = A$.

Which one of the following set of rules will make $L(G) = L(M)$?

A. $\{A \rightarrow aB, A \rightarrow bA, B \rightarrow bA, B \rightarrow aA, B \rightarrow \epsilon)$

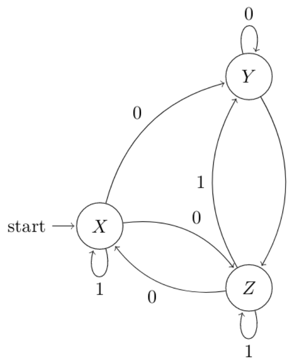
B. $\{A \to aA, A \to bB, B \to aB, B \to bA, B \to \epsilon)$
C. $\{A \to bB, A \to aB, B \to aA, B \to bA, B \to \epsilon)$
D. $\{A \to aA, A \to bA, B \to aB, B \to bA, A \to \epsilon)$

gateit-2004 theory-of-computation finite-automata normal

Answer key☟

6.7.34 Finite Automata: GATE IT 2005 | Question: 37

Consider the non-deterministic finite automaton (NFA) shown in the figure.



State X is the starting state of the automaton. Let the language accepted by the NFA with Y as the only accepting state be $L1$. Similarly, let the language accepted by the NFA with Z as the only accepting state be $L2$. Which of the following statements about $L1$ and $L2$ is TRUE?

A. $L1 = L2$
B. $L1 \subset L2$
C. $L2 \subset L1$
D. None of the above

gateit-2005 theory-of-computation finite-automata normal

Answer key☟

6.7.35 Finite Automata: GATE IT 2005 | Question: 39

Consider the regular grammar:

- $S \to Xa \mid Ya$
- $X \to Za$
- $Z \to Sa \mid \epsilon$
- $Y \to Wa$
- $W \to Sa$

where S is the starting symbol, the set of terminals is $\{a\}$ and the set of non-terminals is $\{S, W, X, Y, Z\}$.
We wish to construct a deterministic finite automaton (DFA) to recognize the same language. What is the minimum number of states required for the DFA?

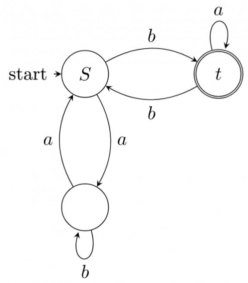
A. 2
B. 3
C. 4
D. 5

gateit-2005 theory-of-computation finite-automata normal

Answer key☟

6.7.36 Finite Automata: GATE IT 2006 | Question: 3

In the automaton below, s is the start state and t is the only final state.



Consider the strings $u = abbaba, v = bab, \text{and} w = aabb$. Which of the following statements is true?

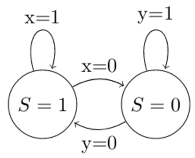
A. The automaton accepts u and v but not w
B. The automaton accepts each of $u, v,$ and w
C. The automaton rejects each of $u, v,$ and w
D. The automaton accepts u but rejects v and w

gateit-2006 theory-of-computation finite-automata easy

Answer key☟

6.7.37 Finite Automata: GATE IT 2006 | Question: 37

For a state machine with the following state diagram the expression for the next state S^+ in terms of the current state S and the input variables x and y is



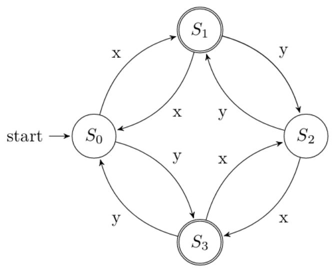
A. $S^+ = S'.y' + S.x$
B. $S^+ = S.x.y' + S'.y.x'$
C. $S^+ = x.y'$
D. $S^+ = S'.y + S.x'$

gateit-2006 theory-of-computation finite-automata normal

Answer key☟

6.7.38 Finite Automata: GATE IT 2007 | Question: 47

Consider the following DFA in which S_0 is the start state and S_1, S_3 are the final states.



What language does this DFA recognize?

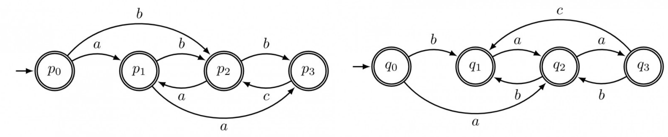
A. All strings of x and y
B. All strings of x and y which have either even number of x and even number of y or odd number of x and odd number of y
C. All strings of x and y which have equal number of x and y
D. All strings of x and y with either even number of x and odd number of y or odd number of x and even number of y

gateit-2007 theory-of-computation finite-automata normal

Answer key☟

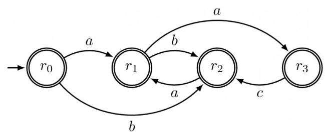
6.7.39 Finite Automata: GATE IT 2007 | Question: 50

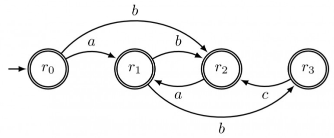
Consider the following finite automata P and Q over the alphabet $\{a, b, c\}$. The start states are indicated by a double arrow and final states are indicated by a double circle. Let the languages recognized by them be denoted by $L(P)$ and $L(Q)$ respectively.

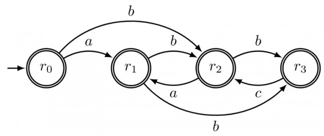


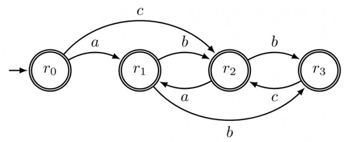
(P) (Q)

The automation which recognizes the language $L(P) \cap L(Q)$ is :

A. 

B. 

C. 

D. 

gateit-2007 theory-of-computation finite-automata normal

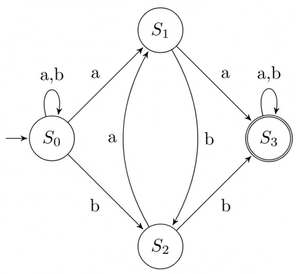
Answer key

6.7.40 Finite Automata: GATE IT 2007 | Question: 71

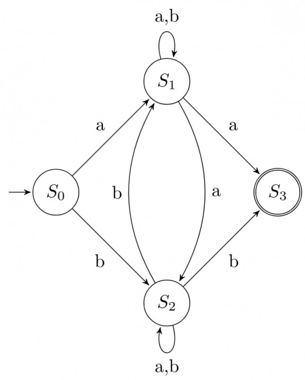
Consider the regular expression $R = (a+b)^*(aa+bb)(a+b)^*$

Which of the following non-deterministic finite automata recognizes the language defined by the regular expression R? Edges labeled λ denote transitions on the empty string.

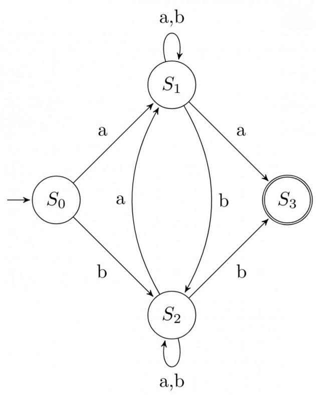
A.



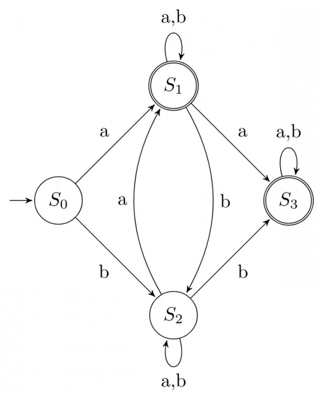
B.



C.



D.



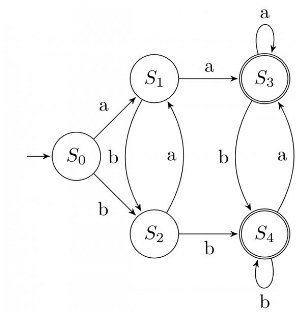
gateit-2007 theory-of-computation finite-automata normal

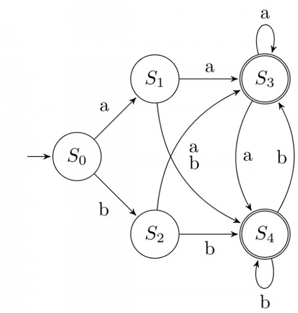
Answer key☟

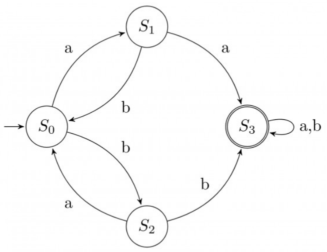
6.7.41 Finite Automata: GATE IT 2007 | Question: 72

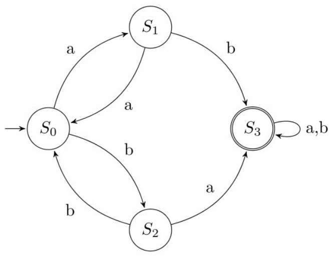
Consider the regular expression $R = (a+b)^*(aa+bb)(a+b)^*$

Which deterministic finite automaton accepts the language represented by the regular expression R?

A. 

B. 

C. 

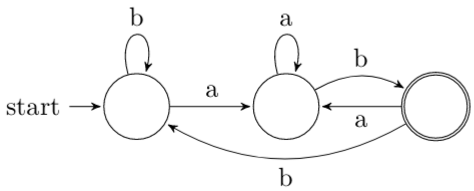
D. 

gateit-2007 theory-of-computation finite-automata normal

Answer key☟

6.7.42 Finite Automata: GATE IT 2008 | Question: 32

If the final states and non-final states in the DFA below are interchanged, then which of the following languages over the alphabet $\{a, b\}$ will be accepted by the new DFA?



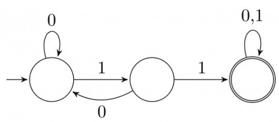
A. Set of all strings that do not end with ab
B. Set of all strings that begin with either an a or $a\ b$
C. Set of all strings that do not contain the substring ab,
D. The set described by the regular expression $b^*aa^*(ba)^*b^*$

gateit-2008 theory-of-computation finite-automata normal

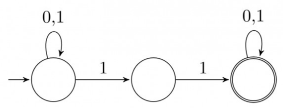
Answer key☟

6.7.43 Finite Automata: GATE IT 2008 | Question: 36

Consider the following two finite automata. M_1 accepts L_1 and M_2 accepts L_2.



M_1



M_2

Which one of the following is TRUE?

A. $L_1 = L_2$
B. $L_1 \subset L_2$
C. $L_1 \cap L_2^C = \varnothing$
D. $L_1 \cup L_2 \neq L_1$

gateit-2008 theory-of-computation finite-automata normal

Answer key

6.8 Finite State Machines (1)

6.8.1 Finite State Machines: GATE CSE 2025 | Set 1 | Question: 49

Consider a finite state machine (FSM) with one input X and one output f, represented by the given state transition table. The minimum number of states required to realize this FSM is ________. (Answer in integer).

Present state	Next state		Output f	
	X=0	X=1	X=0	X=1
A	F	B	0	0
B	D	C	0	0
C	F	E	0	0
D	G	A	1	0
E	D	C	0	0
F	F	B	1	1
G	G	H	0	1
H	G	A	1	0

gatecse2025-set1 theory-of-computation finite-state-machines numerical-answers two-marks

Answer key

6.9 Identify Class Language (31)

✍ **Practice Tests:** Test 1 (15Q) Test 2 (15Q) Test 3 (13Q) Weekly Quiz 8 (10Q)

6.9.1 Identify Class Language: GATE CSE 1987 | Question: 1-xiii

FORTRAN is a:

A. Regular language.
B. Context-free language.
C. Context-senstive language.
D. None of the above.

gate1987 theory-of-computation identify-class-language

Answer key

6.9.2 Identify Class Language: GATE CSE 1988 | Question: 2ix

What is the type of the language L, where $L = \{a^n b^n \mid 0 < n < 327\text{-th prime number}\}$

gate1988 normal descriptive theory-of-computation identify-class-language

Answer key

6.9.3 Identify Class Language: GATE CSE 1991 | Question: 17,a

Show that the Turing machines, which have a read only input tape and constant size work tape, recognize precisely the class of regular languages.

gate1991 theory-of-computation descriptive identify-class-language proof

Answer key

6.9.4 Identify Class Language: GATE CSE 1994 | Question: 19

A. Given a set:

$$S = \{x \mid \text{ there is an x-block of 5's in the decimal expansion of } \pi\}$$

(Note: x-$block$ is a maximal block of x successive 5's)
Which of the following statements is true with respect to S? No reason to be given for the answer.

i. S is regular
ii. S is recursively enumerable
iii. S is not recursively enumerable
iv. S is recursive

B. Given that a language L_1 is regular and that the language $L_1 \cup L_2$ is regular, is the language L_2 always regular? Prove your answer.

gate1994 theory-of-computation identify-class-language normal descriptive

Answer key

6.9.5 Identify Class Language: GATE CSE 1999 | Question: 2.4

If $L1$ is context free language and $L2$ is a regular language which of the following is/are false?

A. $L1 - L2$ is not context free
B. $L1 \cap L2$ is context free
C. $\sim L1$ is context free
D. $\sim L2$ is regular

gate1999 theory-of-computation identify-class-language normal multiple-selects

Answer key

6.9.6 Identify Class Language: GATE CSE 2000 | Question: 1.5

Let L denote the languages generated by the grammar $S \to 0S0 \mid 00.$
Which of the following is TRUE?

A. $L = 0^+$
B. L is regular but not 0^+
C. L is context free but not regular
D. L is not context free

gatecse-2000 theory-of-computation easy identify-class-language

Answer key

6.9.7 Identify Class Language: GATE CSE 2002 | Question: 1.7

The language accepted by a Pushdown Automaton in which the stack is limited to 10 items is best described as

A. Context free
B. Regular
C. Deterministic Context free
D. Recursive

gatecse-2002 theory-of-computation easy identify-class-language

Answer key

6.9.8 Identify Class Language: GATE CSE 2004 | Question: 87

The language $\{a^m b^n c^{m+n} \mid m, n \geq 1\}$ is

A. regular
B. context-free but not regular
C. context-sensitive but not context free
D. type-0 but not context sensitive

gatecse-2004 theory-of-computation normal identify-class-language

Answer key☟

6.9.9 Identify Class Language: GATE CSE 2005 | Question: 55

Consider the languages:

$L_1 = \{a^n b^n c^m \mid n, m > 0\}$ and $L_2 = \{a^n b^m c^m \mid n, m > 0\}$

Which one of the following statements is FALSE?

A. $L_1 \cap L_2$ is a context-free language
B. $L_1 \cup L_2$ is a context-free language
C. $L_1 \text{ and } L_2$ are context-free languages
D. $L_1 \cap L_2$ is a context sensitive language

gatecse-2005 theory-of-computation identify-class-language normal

Answer key☟

6.9.10 Identify Class Language: GATE CSE 2006 | Question: 30

For $s \in (0+1)^*$ let $d(s)$ denote the decimal value of s(e.g. $d(101) = 5)$. Let

$$L = \{s \in (0+1)^* \mid d(s) \bmod 5 = 2 \text{ and } d(s) \bmod 7 \neq 4\}$$

Which one of the following statements is true?

A. L is recursively enumerable, but not recursive
B. L is recursive, but not context-free
C. L is context-free, but not regular
D. L is regular

gatecse-2006 theory-of-computation normal identify-class-language

Answer key☟

6.9.11 Identify Class Language: GATE CSE 2006 | Question: 33

Let L_1 be a regular language, L_2 be a deterministic context-free language and L_3 a recursively enumerable, but not recursive, language. Which one of the following statements is false?

A. $L_1 \cap L_2$ is a deterministic CFL
B. $L_3 \cap L_1$ is recursive
C. $L_1 \cup L_2$ is context free
D. $L_1 \cap L_2 \cap L_3$ is recursively enumerable

gatecse-2006 theory-of-computation normal identify-class-language

Answer key☟

6.9.12 Identify Class Language: GATE CSE 2007 | Question: 30

The language $L = \{0^i 2 1^i \mid i \geq 0\}$ over the alphabet $\{0, 1, 2\}$ is:

A. not recursive
B. is recursive and is a deterministic CFL
C. is a regular language
D. is not a deterministic CFL but a CFL

gatecse-2007 theory-of-computation normal identify-class-language

Answer key☟

6.9.13 Identify Class Language: GATE CSE 2008 | Question: 51

Match the following:

E.	Checking that identifiers are declared before their use	P.	$L = \{a^n b^m c^n d^m \mid n \geq 1, m \geq 1\}$
F.	Number of formal parameters in the declaration of a function agrees with the number of actual parameters in a use of that function	Q.	$X \to XbX \mid XcX \mid dXf \mid g$
G.	Arithmetic expressions with matched pairs of parentheses	R.	$L = \{wcw \mid w \in (a \mid b)^*\}$
H.	Palindromes	S.	$X \to bXb \mid cXc \mid \epsilon$

A. E-P, F-R, G-Q, H-S
B. E-R, F-P, G-S, H-Q
C. E-R, F-P, G-Q, H-S
D. E-P, F-R, G-S, H-Q

gatecse-2008 normal theory-of-computation identify-class-language match-the-following

Answer key

6.9.14 Identify Class Language: GATE CSE 2008 | Question: 9

Which of the following is true for the language

$$\{a^p \mid p \text{ is a prime }\}?$$

A. It is not accepted by a Turing Machine
B. It is regular but not context-free
C. It is context-free but not regular
D. It is neither regular nor context-free, but accepted by a Turing machine

gatecse-2008 theory-of-computation easy identify-class-language

Answer key

6.9.15 Identify Class Language: GATE CSE 2009 | Question: 40

Let $L = L_1 \cap L_2$, where L_1 and L_2 are languages as defined below:

$L_1 = \{a^m b^m c a^n b^n \mid m, n \geq 0\}$

$L_2 = \left\{a^i b^j c^k \mid i, j, k \geq 0\right\}$

Then L is

A. Not recursive
B. Regular
C. Context free but not regular
D. Recursively enumerable but not context free.

gatecse-2009 theory-of-computation easy identify-class-language

Answer key

6.9.16 Identify Class Language: GATE CSE 2010 | Question: 40

Consider the languages

$L1 = \{0^i 1^j \mid i \neq j\}$,

$L2 = \{0^i 1^j \mid i = j\}$,

$L3 = \{0^i 1^j \mid i = 2j + 1\}$,

$L4 = \{0^i 1^j \mid i \neq 2j\}$

A. Only $L2$ is context free.
B. Only $L2$ and $L3$ are context free.
C. Only $L1$ and $L2$ are context free.
D. All are context free

gatecse-2010 theory-of-computation context-free-language identify-class-language normal

Answer key

6.9.17 Identify Class Language: GATE CSE 2011 | Question: 26

Consider the languages $L1,\ L2\,and\,L3$ as given below.

$L1 = \{0^p1^q \mid p, q \in N\},$

$L2 = \{0^p1^q \mid p, q \in N\, and\, p = q\}\, and,$

$L3 = \{0^p1^q0^r \mid p, q, r \in N\, and\, p = q = r\}.$

Which of the following statements is **NOT TRUE**?

A. Push Down Automata (PDA) can be used to recognize $L1$ and $L2$
B. $L1$ is a regular language
C. All the three languages are context free
D. Turing machines can be used to recognize all the languages

gatecse-2011 theory-of-computation identify-class-language normal

Answer key

6.9.18 Identify Class Language: GATE CSE 2013 | Question: 32

Consider the following languages.

$L_1 = \{0^p1^q0^r \mid p, q, r \geq 0\}$

$L_2 = \{0^p1^q0^r \mid p, q, r \geq 0, p \neq r\}$

Which one of the following statements is **FALSE**?

A. L_2 is context-free.
B. $L_1 \cap L_2$ is context-free.
C. Complement of L_2 is recursive.
D. Complement of L_1 is context-free but not regular.

gatecse-2013 theory-of-computation identify-class-language normal

Answer key

6.9.19 Identify Class Language: GATE CSE 2014 | Set 3 | Question: 36

Consider the following languages over the alphabet $\sum = \{0, 1, c\}$

$L_1 = \{0^n1^n \mid n \geq 0\}$

$L_2 = \{wcw^r \mid w \in \{0, 1\}^*\}$

$L_3 = \{ww^r \mid w \in \{0, 1\}^*\}$

Here, w^r is the reverse of the string w. Which of these languages are deterministic Context-free languages?

A. None of the languages
B. Only L_1
C. Only L_1 and L_2
D. All the three languages

gatecse-2014-set3 theory-of-computation identify-class-language context-free-language normal

Answer key

6.9.20 Identify Class Language: GATE CSE 2017 | Set 1 | Question: 37

Consider the context-free grammars over the alphabet $\{a, b, c\}$ given below. S and T are non-terminals.

$G_1 : S \to aSb \mid T, T \to cT \mid \epsilon$

$G_2 : S \to bSa \mid T, T \to cT \mid \epsilon$

The language $L\left(G_1\right) \cap L(G_2)$ is

A. Finite
B. Not finite but regular
C. Context-Free but not regular
D. Recursive but not context-free

gatecse-2017-set1 theory-of-computation context-free-language identify-class-language normal

Answer key

6.9.21 Identify Class Language: GATE CSE 2017 | Set 2 | Question: 40

Consider the following languages.

- $L_1 = \{a^p \mid p \text{ is a prime number}\}$
- $L_2 = \{a^n b^m c^{2m} \mid n \geq 0, m \geq 0\}$
- $L_3 = \{a^n b^n c^{2n} \mid n \geq 0\}$
- $L_4 = \{a^n b^n \mid n \geq 1\}$

Which of the following are CORRECT?

I. L_1 is context free but not regular
II. L_2 is not context free
III. L_3 is not context free but recursive
IV. L_4 is deterministic context free

A. I, II and IV only
B. II and III only
C. I and IV only
D. III and IV only

gatecse-2017-set2 theory-of-computation identify-class-language

Answer key☟

6.9.22 Identify Class Language: GATE CSE 2018 | Question: 35

Consider the following languages:

I. $\{a^m b^n c^p d^q \mid m + p = n + q, \text{ where } m, n, p, q \geq 0\}$
II. $\{a^m b^n c^p d^q \mid m = n \text{ and } p = q, \text{ where } m, n, p, q \geq 0\}$
III. $\{a^m b^n c^p d^q \mid m = n = p \text{ and } p \neq q, \text{ where } m, n, p, q \geq 0\}$
IV. $\{a^m b^n c^p d^q \mid mn = p + q, \text{ where } m, n, p, q \geq 0\}$

Which of the above languages are context-free?

A. I and IV only
B. I and II only
C. II and III only
D. II and IV only

gatecse-2018 theory-of-computation identify-class-language context-free-language normal two-marks

Answer key☟

6.9.23 Identify Class Language: GATE CSE 2020 | Question: 10

Consider the language $L = \{a^n \mid n \geq 0\} \cup \{a^n b^n \mid n \geq 0\}$ and the following statements.

I. L is deterministic context-free.
II. L is context-free but not deterministic context-free.
III. L is not $LL(k)$ for any k.

Which of the above statements is/are TRUE?

A. I only
B. II only
C. I and III only
D. III only

gatecse-2020 theory-of-computation identify-class-language one-mark

Answer key☟

6.9.24 Identify Class Language: GATE CSE 2020 | Question: 32

Consider the following languages.

$$L_1 = \{wxyx \mid w, x, y \in (0+1)^+\}$$
$$L_2 = \{xy \mid x, y \in (a+b)^*, \mid x \mid = \mid y \mid, x \neq y\}$$

Which one of the following is TRUE?

A. L_1 is regular and L_2 is context- free.
B. L_1 context- free but not regular and L_2 is context-free.

C. Neither L_1 nor L_2 is context- free.
D. L_1 context- free but L_2 is not context-free.

gatecse-2020 theory-of-computation identify-class-language two-marks

Answer key

6.9.25 Identify Class Language: GATE CSE 2021 | Set 2 | Question: 12

Let L_1 be a regular language and L_2 be a context-free language. Which of the following languages is/are context-free?

A. $L_1 \cap \overline{L_2}$
B. $\overline{\overline{L_1} \cup \overline{L_2}}$
C. $L_1 \cup (L_2 \cup \overline{L_2})$
D. $(L_1 \cap L_2) \cup (\overline{L_1} \cap L_2)$

gatecse-2021-set2 multiple-selects theory-of-computation identify-class-language one-mark

Answer key

6.9.26 Identify Class Language: GATE CSE 2022 | Question: 13

Which of the following statements is/are TRUE?

A. Every subset of a recursively enumerable language is recursive.
B. If a language L and its complement $\overline{L}$ are both recursively enumerable, then L must be recursive.
C. Complement of a context-free language must be recursive.
D. If L_1 and L_2 are regular, then $L_1 \cap L_2$ must be deterministic context-free.

gatecse-2022 theory-of-computation identify-class-language recursive-and-recursively-enumerable-languages multiple-selects one-mark

Answer key

6.9.27 Identify Class Language: GATE CSE 2022 | Question: 37

Consider the following languages:

$L_1 = \{a^n w a^n | w \in \{a, b\}^*\}$

$L_2 = \{w x w^R | w, x \in \{a, b\}^*, |w|, |x| > 0\}$

Note that w^R is the reversal of the string w. Which of the following is/are TRUE?

A. L_1 and L_2 are regular.
B. L_1 and L_2 are context-free.
C. L_1 is regular and L_2 is context-free.
D. L_1 and L_2 are context-free but not regular.

gatecse-2022 theory-of-computation identify-class-language context-free-language multiple-selects two-marks

Answer key

6.9.28 Identify Class Language: GATE CSE 2023 | Question: 14

Which of the following statements is/are CORRECT?

A. The intersection of two regular languages is regular.
B. The intersection of two context-free languages is context-free.
C. The intersection of two recursive languages is recursive.
D. The intersection of two recursively enumerable languages is recursively enumerable.

gatecse-2023 theory-of-computation identify-class-language multiple-selects one-mark

Answer key

6.9.29 Identify Class Language: GATE IT 2005 | Question: 4

Let L be a regular language and M be a context-free language, both over the alphabet Σ. Let L^c and M^c denote the complements of L and M respectively. Which of the following statements about the language $L^c \cup M^c$ is TRUE?

A. It is necessarily regular but not necessarily context-free.

B. It is necessarily context-free.

C. It is necessarily non-regular.

D. None of the above

gateit-2005 theory-of-computation normal identify-class-language

Answer key

6.9.30 Identify Class Language: GATE IT 2005 | Question: 6

The language $\{0^n 1^n 2^n \mid 1 \leq n \leq 10^6\}$ is

A. regular

B. context-free but not regular

C. context-free but its complement is not context-free

D. not context-free

gateit-2005 theory-of-computation easy identify-class-language

Answer key

6.9.31 Identify Class Language: GATE IT 2008 | Question: 33

Consider the following languages.

- $L_1 = \{a^i b^j c^k \mid i = j, k \geq 1\}$
- $L_2 = \{a^i b^j \mid j = 2i, i \geq 0\}$

Which of the following is true?

A. L_1 is not a CFL but L_2 is

B. $L_1 \cap L_2 = \varnothing$ and L_1 is non-regular

C. $L_1 \cup L_2$ is not a CFL but L_2 is

D. There is a 4-state PDA that accepts L_1, but there is no DPDA that accepts L_2.

gateit-2008 theory-of-computation normal identify-class-language

Answer key

6.10 Medium (1)

6.10.1 Medium: GATE CSE 2006 | Question: 29

If s is a string over $(0+1)^*$ then let $n_0(s)$ denote the number of 0's in s and $n_1(s)$ the number of 1's in s. Which one of the following languages is not regular?

A. $L = \{s \in (0+1)^* \mid n_0(s) \text{ is a 3-digit prime }\}$

B. $L = \{s \in (0+1)^* \mid \text{ for every prefix s' of s, } \mid n_0(s') - n_1(s') \mid \leq 2\}$

C. $L = \{s \in (0+1)^* \mid n_0(s) - n_1(s) \mid \leq 4\}$

D. $L = \{s \in (0+1)^* \mid n_0(s) \mod 7 = n_1(s) \mod 5 = 0\}$

gatecse-2006 theory-of-computation medium regular-language

Answer key

6.11 Minimal State Automata (25)

✍ **Practice Tests:** Test 1 (15Q) Test 2 (15Q)

6.11.1 Minimal State Automata: GATE CSE 1987 | Question: 2j

State whether the following statements are TRUE or FALSE:

A minimal DFA that is equivalent to an NDFA with n nodes has always 2^n states.

gate1987 theory-of-computation finite-automata minimal-state-automata

Answer key

6.11.2 Minimal State Automata: GATE CSE 1997 | Question: 20

Construct a finite state machine with minimum number of states, accepting all strings over (a, b) such that the number of a's is divisible by two and the number of b's is divisible by three.

gate1997 theory-of-computation finite-automata normal minimal-state-automata descriptive

Answer key

6.11.3 Minimal State Automata: GATE CSE 1997 | Question: 70

Following is a state table for time finite state machine.

Present State	Next State Output	
	Input- 0	Input-1
A	B.1	H.1
B	F.1	D.1
C	D.0	E.1
D	C.0	F.1
E	D.1	C.1
F	C.1	C.1
G	C.1	D.1
H	C.0	A.1

A. Find the equivalence partition on the states of the machine.
B. Give the state table for the minimal machine. (Use appropriate names for the equivalent states. For example if states X and Y are equivalent then use XY as the name for the equivalent state in the minimal machine).

gate1997 theory-of-computation minimal-state-automata descriptive

Answer key

6.11.4 Minimal State Automata: GATE CSE 1998 | Question: 2.5

Let L be the set of all binary strings whose last two symbols are the same. The number of states in the minimal state deterministic finite state automaton accepting L is

A. 2
B. 5
C. 8
D. 3

gate1998 theory-of-computation finite-automata normal minimal-state-automata

Answer key

6.11.5 Minimal State Automata: GATE CSE 1998 | Question: 4

Design a deterministic finite state automaton (using minimum number of states) that recognizes the following language:

$L = \{w \in \{0, 1\}^* \mid w$ interpreted as binary number (ignoring the leading zeros) is divisible by five$\}$.

gate1998 theory-of-computation finite-automata normal minimal-state-automata descriptive

Answer key

6.11.6 Minimal State Automata: GATE CSE 1999 | Question: 1.4

Consider the regular expression $(0 + 1)(0 + 1) \dots N$ times. The minimum state finite automaton that recognizes the language represented by this regular expression contains

A. n states
B. $n + 1$ states
C. $n + 2$ states
D. None of the above

gate1999 theory-of-computation finite-automata easy minimal-state-automata

Answer key

6.11.7 Minimal State Automata: GATE CSE 2001 | Question: 1.6

Given an arbitrary non-deterministic finite automaton (NFA) with N states, the maximum number of states in an equivalent minimized DFA at least

A. N^2 B. 2^N C. $2N$ D. $N!$

gatecse-2001 finite-automata theory-of-computation easy minimal-state-automata

Answer key

6.11.8 Minimal State Automata: GATE CSE 2001 | Question: 2.5

Consider a DFA over $\Sigma = \{a, b\}$ accepting all strings which have number of a's divisible by 6 and number of b's divisible by 8. What is the minimum number of states that the DFA will have?

A. 8 B. 14 C. 15 D. 48

gatecse-2001 theory-of-computation finite-automata minimal-state-automata

Answer key

6.11.9 Minimal State Automata: GATE CSE 2002 | Question: 2.13

The smallest finite automaton which accepts the language $\{x|$ length of x is divisible by $3\}$ has

A. 2 states B. 3 states C. 4 states D. 5 states

gatecse-2002 theory-of-computation normal finite-automata minimal-state-automata

Answer key

6.11.10 Minimal State Automata: GATE CSE 2006 | Question: 34

Consider the regular language $L = (111 + 11111)^*$. The minimum number of states in any DFA accepting this languages is:

A. 3 B. 5 C. 8 D. 9

gatecse-2006 theory-of-computation finite-automata normal minimal-state-automata

Answer key

6.11.11 Minimal State Automata: GATE CSE 2007 | Question: 29

A minimum state deterministic finite automaton accepting the language

$L = \{w \mid w \in \{0, 1\}^*,$ number of 0s and 1s in w are divisible by 3 and 5, respectively $\}$ has

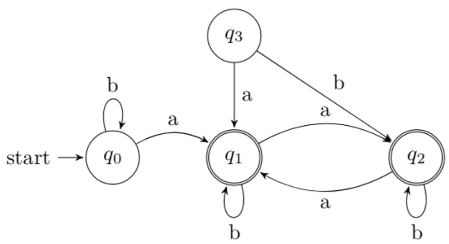
A. 15 states B. 11 states C. 10 states D. 9 states

gatecse-2007 theory-of-computation finite-automata normal minimal-state-automata

Answer key

6.11.12 Minimal State Automata: GATE CSE 2007 | Question: 75

Consider the following Finite State Automaton:



The minimum state automaton equivalent to the above FSA has the following number of states:

A. 1 B. 2 C. 3 D. 4

normal gatecse-2007 theory-of-computation finite-automata minimal-state-automata

Answer key

6.11.13 Minimal State Automata: GATE CSE 2010 | Question: 41

Let w be any string of length n in $\{0,1\}^*$. Let L be the set of all substrings of w. What is the minimum number of states in non-deterministic finite automation that accepts L?

A. $n-1$
B. n
C. $n+1$
D. 2^{n-1}

gatecse-2010 theory-of-computation finite-automata normal minimal-state-automata

Answer key

6.11.14 Minimal State Automata: GATE CSE 2011 | Question: 42

Definition of a language L with alphabet $\{a\}$ is given as following.

$$L = \left\{a^{nk} \mid k > 0,\ \ and\ \ n \text{ is a positive integer constant}\right\}$$

What is the minimum number of states needed in a DFA to recognize L?

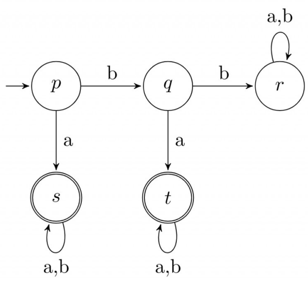
A. $k+1$
B. $n+1$
C. 2^{n+1}
D. 2^{k+1}

gatecse-2011 theory-of-computation finite-automata normal minimal-state-automata

Answer key

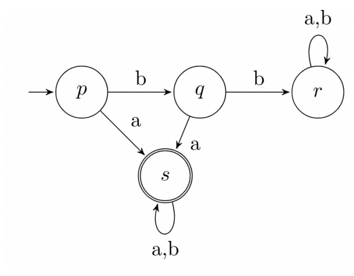
6.11.15 Minimal State Automata: GATE CSE 2011 | Question: 45

A deterministic finite automaton (DFA) D with alphabet $\Sigma = \{a, b\}$ is given below.

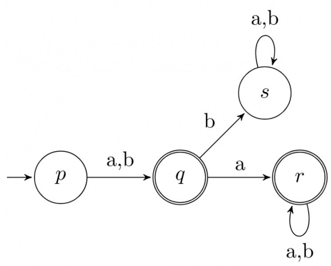


Which of the following finite state machines is a valid minimal DFA which accepts the same languages as D?

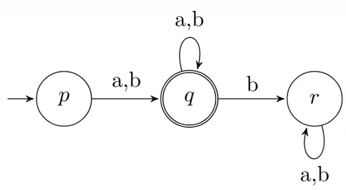
A.



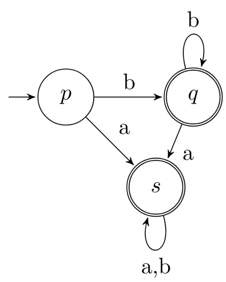
B.



C.



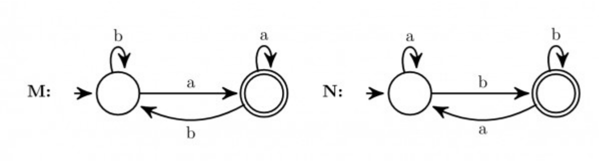
D.



gatecse-2011 theory-of-computation finite-automata easy minimal-state-automata

Answer key

6.11.16 Minimal State Automata: GATE CSE 2015 | Set 1 | Question: 52



Consider the DFAs M and N given above. The number of states in a minimal DFA that accept the language $L(M) \cap L(N)$ is____________.

gatecse-2015-set1 theory-of-computation finite-automata easy numerical-answers minimal-state-automata

Answer key

6.11.17 Minimal State Automata: GATE CSE 2015 | Set 2 | Question: 53

The number of states in the minimal deterministic finite automaton corresponding to the regular expression $(0+1)^*(10)$ is _____.

gatecse-2015-set2 theory-of-computation finite-automata normal numerical-answers minimal-state-automata

Answer key

6.11.18 Minimal State Automata: GATE CSE 2015 | Set 3 | Question: 18

Let L be the language represented by the regular expression $\Sigma^*0011\Sigma^*$ where $\Sigma = \{0, 1\}$. What is the minimum number of states in a DFA that recognizes $\bar{L}$ (complement of L)?

A. 4 B. 5 C. 6 D. 8

gatecse-2015-set3 theory-of-computation finite-automata normal minimal-state-automata

Answer key

6.11.19 Minimal State Automata: GATE CSE 2016 | Set 2 | Question: 16

The number of states in the minimum sized DFA that accepts the language defined by the regular expression.

$$(0+1)^*(0+1)(0+1)^*$$

is ________.

gatecse-2016-set2 theory-of-computation finite-automata normal numerical-answers minimal-state-automata

Answer key

6.11.20 Minimal State Automata: GATE CSE 2017 | Set 1 | Question: 22

Consider the language L given by the regular expression $(a+b)^*b(a+b)$ over the alphabet $\{a, b\}$. The smallest number of states needed in a deterministic finite-state automaton (DFA) accepting L is ___________ .

gatecse-2017-set1 theory-of-computation finite-automata numerical-answers minimal-state-automata

Answer key

6.11.21 Minimal State Automata: GATE CSE 2017 | Set 2 | Question: 25

The minimum possible number of states of a deterministic finite automaton that accepts the regular language L = $\{w_1aw_2 \mid w_1, w_2 \in \{a, b\}^*, |w_1| = 2, |w_2| \geq 3\}$ is ______________ .

theory-of-computation gatecse-2017-set2 finite-automata numerical-answers minimal-state-automata

Answer key

6.11.22 Minimal State Automata: GATE CSE 2018 | Question: 6

Let N be an NFA with n states. Let k be the number of states of a minimal DFA which is equivalent to N. Which one of the following is necessarily true?

A. $k \geq 2^n$
B. $k \geq n$
C. $k \leq n^2$
D. $k \leq 2^n$

gatecse-2018 theory-of-computation minimal-state-automata normal one-mark

Answer key

6.11.23 Minimal State Automata: GATE CSE 2019 | Question: 48

Let Σ be the set of all bijections from $\{1, \dots, 5\}$ to $\{1, \dots, 5\}$, where id denotes the identity function, i.e. $id(j) = j, \forall j$. Let $\circ$ denote composition on functions. For a string $x = x_1 x_2 \dots x_n \in \Sigma^n, n \geq 0$, let $\pi(x) = x_1 \circ x_2 \circ \cdots \circ x_n$. Consider the language $L = \{x \in \Sigma^* \mid \pi(x) = id\}$. The minimum number of states in any DFA accepting L is _______

gatecse-2019 numerical-answers theory-of-computation finite-automata minimal-state-automata difficult two-marks

Answer key

6.11.24 Minimal State Automata: GATE CSE 2023 | Question: 53

Consider the language L over the alphabet $\{0, 1\}$, given below:

$$L = \{w \in \{0,1\}^* \mid w \text{ does not contain three or more consecutive } 1\text{'s}\}.$$

The minimum number of states in a Deterministic Finite-State Automaton (DFA) for L is ___________.

gatecse-2023 theory-of-computation minimal-state-automata numerical-answers two-marks

Answer key

6.11.25 Minimal State Automata: GATE IT 2008 | Question: 6

Let N be an NFA with n states and let M be the minimized DFA with m states recognizing the same language. Which of the following in NECESSARILY true?

A. $m \leq 2^n$
B. $n \leq m$
C. M has one accept state
D. $m = 2^n$

gateit-2008 theory-of-computation finite-automata normal minimal-state-automata

Answer key

6.12 Non Determinism (6)

✍ **Practice Test:** Test 1 (6Q)

6.12.1 Non Determinism: GATE CSE 1992 | Question: 02,xx

In which of the cases stated below is the following statement true?
"For every non-deterministic machine M_1 there exists an equivalent deterministic machine M_2 recognizing the same language".

A. M_1 is non-deterministic finite automaton.
B. M_1 is non-deterministic PDA.
C. M_1 is a non-deterministic Turing machine.
D. For no machines M_1 and M_2, the above statement true.

gate1992 theory-of-computation easy non-determinism multiple-selects

Answer key

6.12.2 Non Determinism: GATE CSE 1994 | Question: 1.16

Which of the following conversions is not possible (algorithmically)?

A. Regular grammar to context free grammar
B. Non-deterministic FSA to deterministic FSA
C. Non-deterministic PDA to deterministic PDA
D. Non-deterministic Turing machine to deterministic Turing machine

gate1994 theory-of-computation easy non-determinism

Answer key

6.12.3 Non Determinism: GATE CSE 1998 | Question: 1.11

Regarding the power of recognition of languages, which of the following statements is false?

A. The non-deterministic finite-state automata are equivalent to deterministic finite-state automata.
B. Non-deterministic Push-down automata are equivalent to deterministic Push-down automata.
C. Non-deterministic Turing machines are equivalent to deterministic Turing machines.
D. Multi-tape Turing machines are available are equivalent to Single-tape Turing machines.

gate1998 theory-of-computation easy non-determinism

Answer key

6.12.4 Non Determinism: GATE CSE 2005 | Question: 54

Let N_f and N_p denote the classes of languages accepted by non-deterministic finite automata and non-deterministic push-down automata, respectively. Let D_f and D_p denote the classes of languages accepted by deterministic finite automata and deterministic push-down automata respectively. Which one of the following is TRUE?

A. $D_f \subset N_f \text{ and } D_p \subset N_p$
B. $D_f \subset N_f \text{ and } D_p = N_p$
C. $D_f = N_f \text{ and } D_p = N_p$
D. $D_f = N_f \text{ and } D_p \subset N_p$

gatecse-2005 theory-of-computation easy non-determinism

Answer key

6.12.5 Non Determinism: GATE CSE 2011 | Question: 8

Which of the following pairs have **DIFFERENT** expressive power?

A. Deterministic finite automata (DFA) and Non-deterministic finite automata (NFA)
B. Deterministic push down automata (DPDA) and Non-deterministic push down automata (NPDA)
C. Deterministic single tape Turing machine and Non-deterministic single tape Turing machine
D. Single tape Turing machine and multi-tape Turing machine

gatecse-2011 theory-of-computation easy non-determinism

Answer key

6.12.6 Non Determinism: GATE IT 2004 | Question: 9

Which one of the following statements is FALSE?

A. There exist context-free languages such that all the context-free grammars generating them are ambiguous
B. An unambiguous context-free grammar always has a unique parse tree for each string of the language generated by it
C. Both deterministic and non-deterministic pushdown automata always accept the same set of languages
D. A finite set of string from some alphabet is always a regular language

gateit-2004 theory-of-computation easy non-determinism

Answer key

6.13 Number of States (2)

6.13.1 Number of States: GATE CSE 2025 | Set 2 | Question: 50

Let $\Sigma = \{1, 2, 3, 4\}$. For $x \in \Sigma^*$, let $\text{prod}(x)$ be the product of symbols in x modulo 7. We take $\text{prod}(\epsilon) = 1$, where ϵ is the null string.

For example, $\text{prod}(124) = (1 \times 2 \times 4) \bmod 7 = 1$.

Define $L = \{x \in \Sigma^* \mid \text{prod}(x) = 2\}$.

The number of states in a minimum state DFA for L is __________ . (Answer in integer)

gatecse2025-set2 theory-of-computation finite-automata number-of-states numerical-answers two-marks

Answer key

6.13.2 Number of States: GATE CSE 2026 | Set 1 | Question: 16

Let M be a nondeterministic finite automaton (NFA) with 6 states over a finite alphabet.

Which of the following options CANNOT be the number of states in the minimal deterministic finite automaton (DFA) that is equivalent to M ?

A. 32
B. 65
C. 1
D. 128

gatecse-2026-set1 theory-of-computation finite-automata number-of-states multiple-selects one-mark

Answer key

6.14 Pumping Lemma (2)

6.14.1 Pumping Lemma: GATE CSE 2019 | Question: 15

For $\Sigma = \{a, b\}$, let us consider the regular language $L = \{x \mid x = a^{2+3k} \text{ or } x = b^{10+12k}, k \geq 0\}$. Which one of the following can be a pumping length (the constant guaranteed by the pumping lemma) for L ?

A. 3
B. 5
C. 9
D. 24

gatecse-2019 theory-of-computation pumping-lemma one-mark

Answer key

6.14.2 Pumping Lemma: GATE IT 2005 | Question: 40

A language L satisfies the Pumping Lemma for regular languages, and also the Pumping Lemma for context-free languages. Which of the following statements about L is TRUE?

A. L is necessarily a regular language.
B. L is necessarily a context-free language, but not necessarily a regular language.
C. L is necessarily a non-regular language.
D. None of the above

gateit-2005 theory-of-computation pumping-lemma easy

Answer key

6.15 Pushdown Automata (15)

✍ **Practice Tests:** Test 1 (15Q) Test 2 (4Q)

6.15.1 Pushdown Automata: GATE CSE 1996 | Question: 13

Let $Q = (\{q_1, q_2\}, \{a, b\}, \{a, b, \perp\}, \delta, \perp, \phi)$ be a pushdown automaton accepting by empty stack for the language which is the set of all nonempty even palindromes over the set $\{a, b\}$. Below is an incomplete specification of the transitions δ. Complete the specification. The top of the stack is assumed to be at the right end of the string representing stack contents.

1. $\delta(q_1, a, \perp) = \{(q_1, \perp a)\}$
2. $\delta(q_1, b, \perp) = \{(q_1, \perp b)\}$
3. $\delta(q_1, a, a) = \{(q_1, aa)\}$
4. $\delta(q_1, b, a) = \{(q_1, ab)\}$
5. $\delta(q_1, a, b) = \{(q_1, ba)\}$
6. $\delta(q_1, b, b) = \{(q_1, bb)\}$
7. $\delta(q_1, a, a) = \{(\dots, \dots)\}$
8. $\delta(q_1, b, b) = \{(\dots, \dots)\}$
9. $\delta(q_2, a, a) = \{(q_2, \epsilon)\}$
10. $\delta(q_2, b, b) = \{(q_2, \epsilon)\}$
11. $\delta(q_2, \epsilon, \perp) = \{(q_2, \epsilon)\}$

gate1996 theory-of-computation pushdown-automata normal descriptive

Answer key

6.15.2 Pushdown Automata: GATE CSE 1997 | Question: 6.6

Which of the following languages over $\{a, b, c\}$ is accepted by a deterministic pushdown automata?

A. $\{wcw^R \mid w \in \{a,b\}^*\}$
B. $\{ww^R \mid w \in \{a,b,c\}^*\}$
C. $\{a^n b^n c^n \mid n \geq 0\}$
D. $\{w \mid w \text{ is a palindrome over } \{a,b,c\}\}$

Note: w^R is the string obtained by reversing $'w'$.

gate1997 theory-of-computation pushdown-automata easy

Answer key

6.15.3 Pushdown Automata: GATE CSE 1998 | Question: 13

Let $M = (\{q_0, q_1\}, \{0, 1\}, \{z_0, X\}, \delta, q_0, z_0, \phi)$ be a Pushdown automation where δ is given by

$\delta(q_0, 1, z_0) = \{(q_0, Xz_0)\}$

$\delta(q_0, \epsilon, z_0) = \{(q_0, \epsilon)\}$

$\delta(q_0, 1, X) = \{(q_0, XX)\}$

$\delta(q_1, 1, X) = \{(q_1, \epsilon)\}$

$\delta(q_0, 0, X) = \{(q_1, X)\}$

$\delta(q_0, 0, z_0) = \{(q_0, z_0)\}$

a. What is the language accepted by this PDA by empty stack?
b. Describe informally the working of the PDA

gate1998 theory-of-computation pushdown-automata descriptive

Answer key

6.15.4 Pushdown Automata: GATE CSE 1999 | Question: 1.6

Let L_1 be the set of all languages accepted by a PDA by final state and L_2 the set of all languages accepted by empty stack. Which of the following is true?

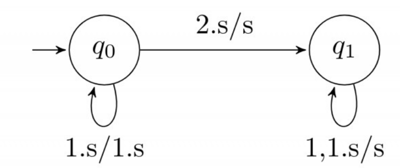
A. $L_1 = L_2$
B. $L_1 \supset L_2$
C. $L_1 \subset L_2$
D. None

normal theory-of-computation gate1999 pushdown-automata

Answer key

6.15.5 Pushdown Automata: GATE CSE 2000 | Question: 8

A push down automation (pda) is given in the following extended notation of finite state diagram:



The nodes denote the states while the edges denote the moves of the pda. The edge labels are of the form $d,\ s/s'$ where d is the input symbol read and s, s' are the stack contents before and after the move. For example the edge labeled $1, s/1.s$ denotes the move from state q_0 to q_0 in which the input symbol 1 is read and pushed to the stack.

A. Introduce two edges with appropriate labels in the above diagram so that the resulting pda accepts the language $\left\{x2x^R \mid x \in \{0,1\}^*, x^R \text{ denotes reverse of x}\right\}$, by empty stack.
B. Describe a non-deterministic pda with three states in the above notation that accept the language $\{0^n1^m \mid n \leq m \leq 2n\}$ by empty stack

gatecse-2000 theory-of-computation descriptive pushdown-automata

Answer key☟

6.15.6 Pushdown Automata: GATE CSE 2001 | Question: 6

Give a deterministic PDA for the language $L = \{a^ncb^{2n} \mid n \geq 1\}$ over the alphabet $\Sigma = \{a, b, c\}$. Specify the acceptance state.

gatecse-2001 theory-of-computation normal pushdown-automata descriptive

Answer key☟

6.15.7 Pushdown Automata: GATE CSE 2009 | Question: 16, ISRO2017-12

Which one of the following is FALSE?

A. There is a unique minimal DFA for every regular language
B. Every NFA can be converted to an equivalent PDA.
C. Complement of every context-free language is recursive.
D. Every nondeterministic PDA can be converted to an equivalent deterministic PDA.

gatecse-2009 theory-of-computation easy isro2017 pushdown-automata

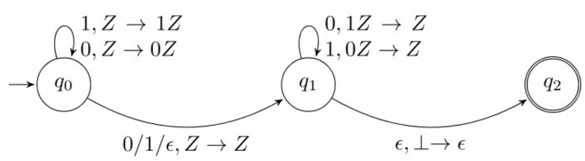
Answer key☟

6.15.8 Pushdown Automata: GATE CSE 2015 | Set 1 | Question: 51

Consider the NPDA

$$\langle Q = \{q_0, q_1, q_2\}, \Sigma = \{0, 1\}, \Gamma = \{0, 1, \perp\}, \delta, q_0, \perp, F = \{q_2\}\rangle$$

, where (as per usual convention) Q is the set of states, Σ is the input alphabet, Γ is the stack alphabet, δ is the state transition function q_0 is the initial state, $\perp$ is the initial stack symbol, and F is the set of accepting states. The state transition is as follows:



Which one of the following sequences must follow the string 101100 so that the overall string is accepted by the automaton?

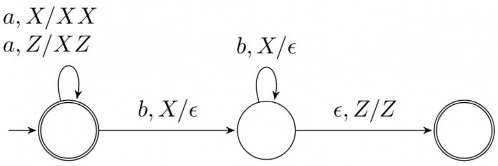
A. 10110 B. 10010 C. 01010 D. 01001

gatecse-2015-set1 theory-of-computation pushdown-automata normal

Answer key

6.15.9 Pushdown Automata: GATE CSE 2016 | Set 1 | Question: 43

Consider the transition diagram of a PDA given below with input alphabet $\Sigma = \{a, b\}$ and stack alphabet $\Gamma = \{X, Z\}$. Z is the initial stack symbol. Let L denote the language accepted by the PDA



Which one of the following is **TRUE**?

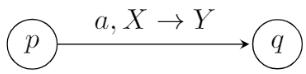
A. $L = \{a^n b^n \mid n \geq 0\}$ and is not accepted by any finite automata
B. $L = \{a^n \mid n \geq 0\} \cup \{a^n b^n \mid n \geq 0\}$ and is not accepted by any deterministic PDA
C. L is not accepted by any Turing machine that halts on every input
D. $L = \{a^n \mid n \geq 0\} \cup \{a^n b^n \mid n \geq 0\}$ and is deterministic context-free

gatecse-2016-set1 theory-of-computation pushdown-automata normal

Answer key

6.15.10 Pushdown Automata: GATE CSE 2021 | Set 1 | Question: 51

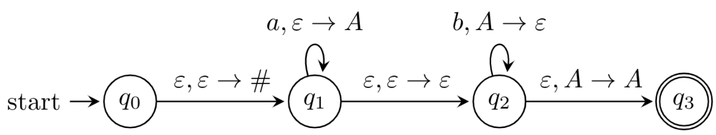
In a pushdown automaton $P = (Q, \Sigma, \Gamma, \delta, q_0, F)$, a transition of the form,



where $p, q \in Q$, $a \in \Sigma \cup \{\epsilon\}$, and $X, Y \in \Gamma \cup \{\epsilon\}$, represents

$$(q, Y) \in \delta(p, a, X).$$

Consider the following pushdown automaton over the input alphabet $\Sigma = \{a, b\}$ and stack alphabet $\Gamma = \{\#, A\}$.



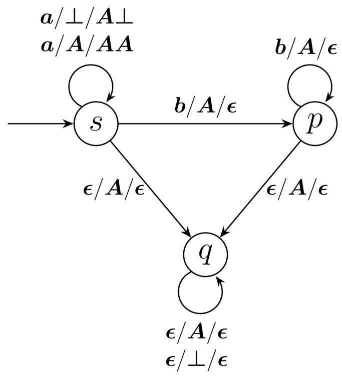
The number of strings of length 100 accepted by the above pushdown automaton is ___________

gatecse-2021-set1 theory-of-computation pushdown-automata numerical-answers two-marks

Answer key

6.15.11 Pushdown Automata: GATE CSE 2023 | Question: 30

Consider the pushdown automaton (PDA) P below, which runs on the input alphabet $\{a, b\}$, has stack alphabet $\{\perp, A\}$, and has three states $\{s, p, q\}$, with s being the start state. A transition from state u to state v, labelled $c/X/\gamma$, where c is an input symbol or ϵ, X is a stack symbol, and γ is a string of stack symbols, represents the fact that in state u, the (PDA) can read c from the input, with X on the top of its stack, pop X from the stack, push in the string γ on the stack, and go to state v. In the initial configuration, the stack has only the symbol $\perp$ in it. The (PDA) accepts by empty stack.



Which one of the following options correctly describes the language accepted by P?

A. $\{a^m b^n \mid 1 \leq m$ and $n < m\}$
B. $\{a^m b^n \mid 0 \leq n \leq m\}$
C. $\{a^m b^n \mid 0 \leq m$ and $0 \leq n\}$
D. $\{a^m \mid 0 \leq m\} \cup \{b^n \mid 0 \leq n\}$

gatecse-2023 theory-of-computation pushdown-automata two-marks

Answer key

6.15.12 Pushdown Automata: GATE IT 2004 | Question: 40

Let $M = (K, \Sigma, \Gamma, \Delta, s, F)$ be a pushdown automaton, where
$K = (s, f), F = \{f\}, \Sigma = \{a, b\}, \Gamma = \{a\}$ and
$\Delta = \{((s, a, \epsilon), (s, a)), ((s, b, \epsilon), (s, a)), ((s, a, a), (f, \epsilon)), ((f, a, a), (f, \epsilon)), ((f, b, a), (f, \epsilon))\}$.
Which one of the following strings is not a member of $L(M)$?

A. aaa B. $aabab$ C. $baaba$ D. bab

gateit-2004 theory-of-computation pushdown-automata normal

Answer key

6.15.13 Pushdown Automata: GATE IT 2005 | Question: 38

Let P be a non-deterministic push-down automaton (NPDA) with exactly one state, q, and exactly one symbol, Z, in its stack alphabet. State q is both the starting as well as the accepting state of the PDA. The stack is initialized with one Z before the start of the operation of the PDA. Let the input alphabet of the PDA be Σ. Let $L(P)$ be the language accepted by the PDA by reading a string and reaching its accepting state. Let $N(P)$ be the language accepted by the PDA by reading a string and emptying its stack.
Which of the following statements is TRUE?

A. $L(P)$ is necessarily Σ^* but $N(P)$ is not necessarily Σ^*.
B. $N(P)$ is necessarily Σ^* but $L(P)$ is not necessarily Σ^*.
C. Both $L(P)$ and $N(P)$ are necessarily Σ^*.
D. Neither $L(P)$ nor $N(P)$ are necessarily Σ^*

gateit-2005 theory-of-computation pushdown-automata normal

Answer key

6.15.14 Pushdown Automata: GATE IT 2006 | Question: 31

Which of the following languages is accepted by a non-deterministic pushdown automaton (PDA) but NOT by a deterministic PDA?

A. $\{a^n b^n c^n \mid n \geq 0\}$
B. $\{a^l b^m c^n \mid l \neq m \text{ or } m \neq n\}$
C. $\{a^n b^n \mid n \geq 0\}$
D. $\{a^m b^n \mid m, n \geq 0\}$

gateit-2006 theory-of-computation pushdown-automata normal

Answer key

6.15.15 Pushdown Automata: GATE IT 2006 | Question: 33

Consider the pushdown automaton (PDA) below which runs over the input alphabet (a, b, c). It has the stack alphabet $\{Z_0, X\}$ where Z_0 is the bottom-of-stack marker. The set of states of the PDA is (s, t, u, f) where s is the start state and f is the final state. The PDA accepts by final state. The transitions of the PDA given below are depicted in a standard manner. For example, the transition $(s, b, X) \rightarrow (t, XZ_0)$ means that if the PDA is in state s and the symbol on the top of the stack is X, then it can read b from the input and move to state t after popping the top of stack and pushing the symbols Z_0 and X (in that order) on the stack.

$(s, a, Z_0) \rightarrow (s, XXZ_0)$
$(s, \epsilon, Z_0) \rightarrow (f, \epsilon)$
$(s, a, X) \rightarrow (s, XXX)$
$(s, b, X) \rightarrow (t, \epsilon)$
$(t, b, X) \rightarrow (t, \epsilon)$
$(t, c, X) \rightarrow (u, \epsilon)$
$(u, c, X) \rightarrow (u, \epsilon)$
$(u, \epsilon, Z_0) \rightarrow (f, \epsilon)$

The language accepted by the PDA is

A. $\{a^l b^m c^n \mid l = m = n\}$
B. $\{a^l b^m c^n \mid l = m\}$
C. $\{a^l b^m c^n \mid 2l = m + n\}$
D. $\{a^l b^m c^n \mid m = n\}$

gateit-2006 theory-of-computation pushdown-automata normal

Answer key

6.16 Recursive and Recursively Enumerable Languages (16)

Practice Tests: Test 1 (15Q) Test 2 (11Q)

6.16.1 Recursive and Recursively Enumerable Languages: GATE CSE 1990 | Question: 3-vi

Recursive languages are:

A. A proper superset of context free languages.
B. Always recognizable by pushdown automata.
C. Also called type 0 languages.
D. Recognizable by Turing machines.

gate1990 normal theory-of-computation turing-machine recursive-and-recursively-enumerable-languages multiple-selects

Answer key

6.16.2 Recursive and Recursively Enumerable Languages: GATE CSE 2003 | Question: 13

Nobody knows yet if $P = NP$. Consider the language L defined as follows.

$$L = \begin{cases} (0+1)^* & \text{if } P = NP \\ \phi & otherwise \end{cases}$$

Which of the following statements is true?

A. L is recursive
B. L is recursively enumerable but not recursive
C. L is not recursively enumerable
D. Whether L is recursively enumerable or not will be known after we find out if $P = NP$

gatecse-2003 theory-of-computation normal recursive-and-recursively-enumerable-languages

Answer key

6.16.3 Recursive and Recursively Enumerable Languages: GATE CSE 2003 | Question: 15

If the strings of a language L can be effectively enumerated in lexicographic (i.e., alphabetic) order, which of the following statements is true?

A. L is necessarily finite
B. L is regular but not necessarily finite
C. L is context free but not necessarily regular
D. L is recursive but not necessarily context-free

theory-of-computation gatecse-2003 normal recursive-and-recursively-enumerable-languages

Answer key

6.16.4 Recursive and Recursively Enumerable Languages: GATE CSE 2003 | Question: 54

Define languages L_0 and L_1 as follows :

- $L_0 = \{\langle M, w, 0 \rangle \mid M \text{ halts on } w\}$
- $L_1 = \{\langle M, w, 1 \rangle \mid M \text{ does not halts on } w\}$

Here $\langle M, w, i \rangle$ is a triplet, whose first component M is an encoding of a Turing Machine, second component w is a string, and third component i is a bit.

Let $L = L_0 \cup L_1$. Which of the following is true?

A. L is recursively enumerable, but L' is not
B. L' is recursively enumerable, but L is not
C. Both L and L' are recursive
D. Neither L nor L' is recursively enumerable

theory-of-computation recursive-and-recursively-enumerable-languages gatecse-2003 difficult

Answer key

6.16.5 Recursive and Recursively Enumerable Languages: GATE CSE 2004 | Question: 89

L_1 is a recursively enumerable language over Σ. An algorithm A effectively enumerates its words as $\omega_1, \omega_2, \omega_3, \dots.$ Define another language L_2 over $\Sigma \cup \{\#\}$ as $\{w_i \# w_j \mid w_i, w_j \in L_1, i < j\}$. Here # is new symbol. Consider the following assertions.

- S_1 : L_1 is recursive implies L_2 is recursive
- S_2 : L_2 is recursive implies L_1 is recursive

Which of the following statements is true?

A. Both S_1 and S_2 are true
B. S_1 is true but S_2 is not necessarily true
C. S_2 is true but S_1 is not necessarily true
D. Neither is necessarily true

gatecse-2004 theory-of-computation recursive-and-recursively-enumerable-languages difficult

Answer key

6.16.6 Recursive and Recursively Enumerable Languages: GATE CSE 2005 | Question: 56

Let L_1 be a recursive language, and let L_2 be a recursively enumerable but not a recursive language. Which one of the following is TRUE?

A. L_1' is recursive and L_2' is recursively enumerable
B. L_1' is recursive and L_2' is not recursively enumerable
C. L_1' and L_2' are recursively enumerable
D. L_1' is recursively enumerable and L_2' is recursive

gatecse-2005 theory-of-computation recursive-and-recursively-enumerable-languages easy

Answer key

6.16.7 Recursive and Recursively Enumerable Languages: GATE CSE 2008 | Question: 13, ISRO2016-36

If L and $\overline{L}$ are recursively enumerable then L is

A. regular
B. context-free
C. context-sensitive
D. recursive

gatecse-2008 theory-of-computation easy isro2016 recursive-and-recursively-enumerable-languages

Answer key

6.16.8 Recursive and Recursively Enumerable Languages: GATE CSE 2008 | Question: 48

Which of the following statements is false?

A. Every NFA can be converted to an equivalent DFA
B. Every non-deterministic Turing machine can be converted to an equivalent deterministic Turing machine
C. Every regular language is also a context-free language
D. Every subset of a recursively enumerable set is recursive

gatecse-2008 theory-of-computation easy recursive-and-recursively-enumerable-languages

Answer key

6.16.9 Recursive and Recursively Enumerable Languages: GATE CSE 2010 | Question: 17

Let L_1 be the recursive language. Let L_2 and L_3 be languages that are recursively enumerable but not recursive. Which of the following statements is not necessarily true?

A. $L_2 - L_1$ is recursively enumerable.
B. $L_1 - L_3$ is recursively enumerable.
C. $L_2 \cap L_3$ is recursively enumerable.
D. $L_2 \cup L_3$ is recursively enumerable.

gatecse-2010 theory-of-computation recursive-and-recursively-enumerable-languages decidability normal

Answer key

6.16.10 Recursive and Recursively Enumerable Languages: GATE CSE 2014 | Set 1 | Question: 35

Let L be a language and $\bar{L}$ be its complement. Which one of the following is **NOT** a viable possibility?

A. Neither L nor $\bar{L}$ is recursively enumerable $(r.e.)$.
B. One of L and $\bar{L}$ is r.e. but not recursive; the other is not r.e.
C. Both L and $\bar{L}$ are r.e. but not recursive.
D. Both L and $\bar{L}$ are recursive.

gatecse-2014-set1 theory-of-computation easy recursive-and-recursively-enumerable-languages

Answer key

6.16.11 Recursive and Recursively Enumerable Languages: GATE CSE 2014 | Set 2 | Question: 16

Let $A \leq_m B$ denotes that language A is mapping reducible (also known as many-to-one reducible) to language B. Which one of the following is FALSE?

A. If $A \leq_m B$ and B is recursive then A is recursive.
B. If $A \leq_m B$ and A is undecidable then B is undecidable.
C. If $A \leq_m B$ and B is recursively enumerable then A is recursively enumerable.
D. If $A \leq_m B$ and B is not recursively enumerable then A is not recursively enumerable.

gatecse-2014-set2 theory-of-computation recursive-and-recursively-enumerable-languages normal

Answer key

6.16.12 Recursive and Recursively Enumerable Languages: GATE CSE 2014 | Set 2 | Question: 35

Let $\langle M\rangle$ be the encoding of a Turing machine as a string over $\Sigma=\{0,1\}$. Let

$$L=\{\langle M\rangle \mid M \text{ is a Turing machine} \\ \text{that accepts a string of length } 2014\}.$$

Then L is:

A. decidable and recursively enumerable
B. undecidable but recursively enumerable
C. undecidable and not recursively enumerable
D. decidable but not recursively enumerable

gatecse-2014-set2 theory-of-computation recursive-and-recursively-enumerable-languages normal

Answer key

6.16.13 Recursive and Recursively Enumerable Languages: GATE CSE 2015 | Set 1 | Question: 3

For any two languages L_1 and L_2 such that L_1 is context-free and L_2 is recursively enumerable but not recursive, which of the following is/are necessarily true?

I. $\bar{L}_1$ (Complement of L_1) is recursive
II. $\bar{L}_2$ (Complement of L_2) is recursive
III. $\bar{L}_1$ is context-free
IV. $\bar{L}_1 \cup L_2$ is recursively enumerable

A. I only
B. III only
C. III and IV only
D. I and IV only

gatecse-2015-set1 theory-of-computation recursive-and-recursively-enumerable-languages normal

Answer key

6.16.14 Recursive and Recursively Enumerable Languages: GATE CSE 2016 | Set 2 | Question: 44

Consider the following languages.

- $L_1=\{\langle M\rangle \mid M \text{ takes at least 2016 steps on some input}\}$,
- $L_2=\{\langle M\rangle \mid M \text{ takes at least 2016 steps on all inputs}\}$ and
- $L_3=\{\langle M\rangle \mid M \text{ accepts } \epsilon\}$,

where for each Turing machine M, $\langle M\rangle$ denotes a specific encoding of M. Which one of the following is TRUE?

A. L_1 is recursive and L_2, L_3 are not recursive
B. L_2 is recursive and L_1, L_3 are not recursive
C. L_1, L_2 are recursive and L_3 is not recursive
D. L_1, L_2, L_3 are recursive

gatecse-2016-set2 theory-of-computation recursive-and-recursively-enumerable-languages

Answer key

6.16.15 Recursive and Recursively Enumerable Languages: GATE CSE 2021 | Set 1 | Question: 12

Let $\langle M\rangle$ denote an encoding of an automaton M. Suppose that $\Sigma=\{0,1\}$. Which of the following languages is/are NOT recursive?

A. $L=\{\langle M\rangle \mid M \text{ is a } \mathrm{DFA} \text{ such that } L(M)=\emptyset\}$
B. $L=\{\langle M\rangle \mid M \text{ is a } \mathrm{DFA} \text{ such that } L(M)=\Sigma^*\}$
C. $L=\{\langle M\rangle \mid M \text{ is a } \mathrm{PDA} \text{ such that } L(M)=\emptyset\}$
D. $L=\{\langle M\rangle \mid M \text{ is a } \mathrm{PDA} \text{ such that } L(M)=\Sigma^*\}$

gatecse-2021-set1 multiple-selects theory-of-computation recursive-and-recursively-enumerable-languages one-mark

Answer key

6.16.16 Recursive and Recursively Enumerable Languages: GATE CSE 2021 | Set 1 | Question: 39

For a Turing machine M, $\langle M \rangle$ denotes an encoding of M. Consider the following two languages.

$$L_1 = \{\langle M \rangle \mid M \text{ takes more than 2021 steps on all inputs}\}$$
$$L_2 = \{\langle M \rangle \mid M \text{ takes more than 2021 steps on some input}\}$$

Which one of the following options is correct?

A. Both L_1 and L_2 are decidable
B. L_1 is decidable and L_2 is undecidable
C. L_1 is undecidable and L_2 is decidable
D. Both L_1 and L_2 are undecidable

gatecse-2021-set1 theory-of-computation recursive-and-recursively-enumerable-languages decidability easy two-marks

Answer key

6.17 Reduction (2)

6.17.1 Reduction: GATE CSE 2005 | Question: 45

Consider three decision problems P_1, P_2 and P_3. It is known that P_1 is decidable and P_2 is undecidable. Which one of the following is TRUE?

A. P_3 is decidable if P_1 is reducible to P_3
B. P_3 is undecidable if P_3 is reducible to P_2
C. P_3 is undecidable if P_2 is reducible to P_3
D. P_3 is decidable if P_3 is reducible to P_2's complement

gatecse-2005 theory-of-computation decidability normal reduction

Answer key

6.17.2 Reduction: GATE CSE 2016 | Set 1 | Question: 44

Let X be a recursive language and Y be a recursively enumerable but not recursive language. Let W and Z be two languages such that $\overline{Y}$ reduces to W, and Z reduces to $\overline{X}$ (reduction means the standard many-one reduction). Which one of the following statements is TRUE?

A. W can be recursively enumerable and Z is recursive.
B. W can be recursive and Z is recursively enumerable.
C. W is not recursively enumerable and Z is recursive.
D. W is not recursively enumerable and Z is not recursive.

gatecse-2016-set1 theory-of-computation easy recursive-and-recursively-enumerable-languages reduction

Answer key

6.18 Regular Expression (29)

✍ **Practice Tests:** Test 1 (15Q) Test 2 (15Q) Test 3 (7Q)

6.18.1 Regular Expression: GATE CSE 1987 | Question: 10d

Give a regular expression over the alphabet $\{0, 1\}$ to denote the set of proper non-null substrings of the string 0110.

gate1987 theory-of-computation regular-expression descriptive

Answer key

6.18.2 Regular Expression: GATE CSE 1991 | Question: 03,xiii

Let $r = 1(1+0)^*, s = 11^*0$ and $t = 1^*0$ be three regular expressions. Which one of the following is true?

A. $L(s) \subseteq L(r)$ and $L(s) \subseteq L(t)$
B. $L(r) \subseteq L(s)$ and $L(s) \subseteq L(t)$
C. $L(s) \subseteq L(t)$ and $L(s) \subseteq L(r)$
D. $L(t) \subseteq L(s)$ and $L(s) \subseteq L(r)$
E. None of the above

gate1991 theory-of-computation regular-expression normal multiple-selects

Answer key

6.18.3 Regular Expression: GATE CSE 1992 | Question: 02,xvii

Which of the following regular expression identities is/are TRUE?

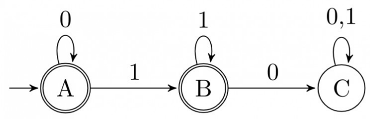
A. $r^{(*)} = r^*$
B. $(r^* s^*) = (r+s)^*$
C. $(r+s)^* = r^* + s^*$
D. $r^* s^* = r^* + s^*$

gate1992 theory-of-computation regular-expression easy

Answer key

6.18.4 Regular Expression: GATE CSE 1994 | Question: 2.10

The regular expression for the language recognized by the finite state automaton of figure is ________



gate1994 theory-of-computation finite-automata regular-expression easy fill-in-the-blanks

Answer key

6.18.5 Regular Expression: GATE CSE 1995 | Question: 1.9 , ISRO2017-13

In some programming language, an identifier is permitted to be a letter followed by any number of letters or digits. If L and D denote the sets of letters and digits respectively, which of the following expressions defines an identifier?

A. $(L+D)^+$
B. $(L.D)^*$
C. $L(L+D)^*$
D. $L(L.D)^*$

gate1995 theory-of-computation regular-expression easy isro2017

Answer key

6.18.6 Regular Expression: GATE CSE 1996 | Question: 1.8

Which two of the following four regular expressions are equivalent? (ε is the empty string).

i. $(00)^*(\varepsilon+0)$
ii. $(00)^*$
iii. 0^*
iv. $0(00)^*$

A. (i) and (ii)
B. (ii) and (iii)
C. (i) and (iii)
D. (iii) and (iv)

gate1996 theory-of-computation regular-expression easy

Answer key

6.18.7 Regular Expression: GATE CSE 1997 | Question: 6.4

Which one of the following regular expressions over $\{0,1\}$ denotes the set of all strings not containing 100 as substring?

A. $0^*(1+0)^*$
B. 0^*1010^*
C. $0^*1^*01^*$
D. $0^*(10+1)^*$

gate1997 theory-of-computation regular-expression normal

Answer key

6.18.8 Regular Expression: GATE CSE 1998 | Question: 1.12

The string 1101 does not belong to the set represented by

A. $110^*(0+1)$
B. $1(0+1)^*101$
C. $(10)^*(01)^*(00+11)^*$
D. $(00+(11)^*0)^*$

gate1998 theory-of-computation regular-expression easy multiple-selects

Answer key

6.18.9 Regular Expression: GATE CSE 1998 | Question: 1.9

If the regular set A is represented by $A=(01+1)^*$ and the regular set B is represented by $B=((01)^*1^*)^*$, which of the following is true?

A. $A \subset B$
B. $B \subset A$
C. A and B are incomparable
D. $A=B$

gate1998 theory-of-computation regular-expression normal

Answer key

6.18.10 Regular Expression: GATE CSE 1998 | Question: 3b

Give a regular expression for the set of binary strings where every 0 is immediately followed by exactly k 1's and preceded by at least k 1's (k is a fixed integer)

gate1998 theory-of-computation regular-expression easy descriptive

Answer key

6.18.11 Regular Expression: GATE CSE 2000 | Question: 1.4

Let S and T be languages over $\Sigma=\{a,b\}$ represented by the regular expressions $(a+b^*)^*$ and $(a+b)^*$, respectively. Which of the following is true?

A. $S \subset T$
B. $T \subset S$
C. $S=T$
D. $S \cap T=\phi$

gatecse-2000 theory-of-computation regular-expression easy

Answer key

6.18.12 Regular Expression: GATE CSE 2003 | Question: 14

The regular expression $0^*(10^*)^*$ denotes the same set as

A. $(1^*0)^*1^*$
B. $0+(0+10)^*$
C. $(0+1)^*10(0+1)^*$
D. None of the above

gatecse-2003 theory-of-computation regular-expression easy

Answer key

6.18.13 Regular Expression: GATE CSE 2009 | Question: 15

Which one of the following languages over the alphabet $\{0,1\}$ is described by the regular expression: $(0+1)^*0(0+1)^*0(0+1)^*$?

A. The set of all strings containing the substring 00
B. The set of all strings containing at most two 0's
C. The set of all strings containing at least two 0's
D. The set of all strings that begin and end with either 0 or 1

gatecse-2009 theory-of-computation regular-expression easy

Answer key

6.18.14 Regular Expression: GATE CSE 2010 | Question: 39

Let $L = \{w \in (0+1)^* \mid w \text{ has even number of } 1s\}$. i.e., L is the set of all the bit strings with even numbers of 1s. Which one of the regular expressions below represents L?

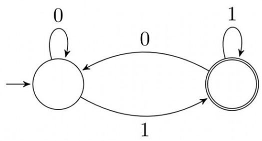
A. $(0^*10^*1)^*$
B. $0^*(10^*10^*)^*$
C. $0^*(10^*1)^*0^*$
D. $0^*1(10^*1)^*10^*$

gatecse-2010 theory-of-computation regular-expression normal

Answer key☟

6.18.15 Regular Expression: GATE CSE 2014 | Set 1 | Question: 36

Which of the regular expressions given below represent the following DFA?



I. $0^*1(1+00^*1)^*$
II. $0^*1^*1+11^*0^*1$
III. $(0+1)^*1$

A. I and II only
B. I and III only
C. II and III only
D. I, II and III

gatecse-2014-set1 theory-of-computation regular-expression finite-automata easy

Answer key☟

6.18.16 Regular Expression: GATE CSE 2014 | Set 3 | Question: 15

The length of the shortest string NOT in the language (over $\Sigma = \{a, b\}$) of the following regular expression is _______.

$$a^*b^*(ba)^*a^*$$

gatecse-2014-set3 theory-of-computation regular-expression numerical-answers easy

Answer key☟

6.18.17 Regular Expression: GATE CSE 2016 | Set 1 | Question: 18

Which one of the following regular expressions represents the language: *the set of all binary strings having two consecutive* 0*'s and two consecutive* 1*'s*?

A. $(0+1)^*0011(0+1)^* + (0+1)^*1100(0+1)^*$
B. $(0+1)^*(00(0+1)^*11 + 11(0+1)^*00)(0+1)^*$
C. $(0+1)^*00(0+1)^* + (0+1)^*11(0+1)^*$
D. $00(0+1)^*11 + 11(0+1)^*00$

gatecse-2016-set1 theory-of-computation regular-expression normal

Answer key☟

6.18.18 Regular Expression: GATE CSE 2020 | Question: 7

Which one of the following regular expressions represents the set of all binary strings with an odd number of $1'$s?

A. $((0+1)^*1(0+1)^*1)^*10^*$
B. $(0^*10^*10^*)^*0^*1$
C. $10^*(0^*10^*10^*)^*$
D. $(0^*10^*10^*)^*10^*$

gatecse-2020 regular-expression normal theory-of-computation one-mark

Answer key☟

6.18.19 Regular Expression: GATE CSE 2021 | Set 2 | Question: 47

Which of the following regular expressions represent(s) the set of all binary numbers that are divisible by three? Assume that the string ϵ is divisible by three.

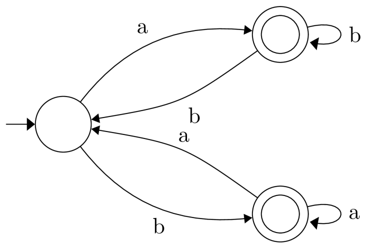
A. $(0+1(01^*0)^*1)^*$
B. $(0+11+10(1+00)^*01)^*$
C. $(0^*(1(01^*0)^*1)^*)^*$
D. $(0+11+11(1+00)^*00)^*$

gatecse-2021-set2 multiple-selects theory-of-computation regular-expression two-marks

Answer key

6.18.20 Regular Expression: GATE CSE 2022 | Question: 2

Which one of the following regular expressions correctly represents the language of the finite automaton given below?



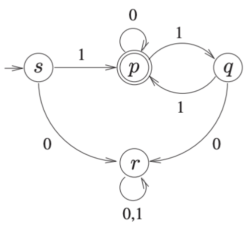
A. $ab^*bab^*+ba^*aba^*$
B. $(ab^*b)^*ab^*+(ba^*a)^*ba^*$
C. $(ab^*b+ba^*a)^*(a^*+b^*)$
D. $(ba^*a+ab^*b)^*(ab^*+ba^*)$

gatecse-2022 theory-of-computation finite-automata regular-expression one-mark

Answer key

6.18.21 Regular Expression: GATE CSE 2023 | Question: 4

Consider the Deterministic Finite-state Automaton (DFA) $\mathcal{A}$ shown below. The DFA runs on the alphabet $\{0,1\}$, and has the set of states $\{s,p,q,r\}$, with s being the start state and p being the only final state.



Which one of the following regular expressions correctly describes the language accepted by $\mathcal{A}$?

A. $1(0^*11)^*$
B. $0(0+1)^*$
C. $1(0+11)^*$
D. $1(110^*)^*$

gatecse-2023 theory-of-computation regular-expression one-mark

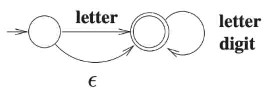
Answer key

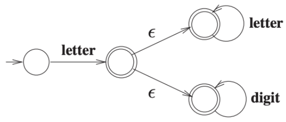
6.18.22 Regular Expression: GATE CSE 2023 | Question: 9

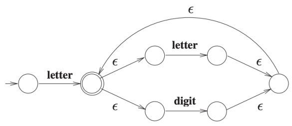
Consider the following definition of a lexical token **id** for an identifier in a programming language, using extended regular expressions:

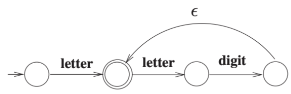
$$\begin{aligned}
&\textbf{letter} && \rightarrow [\mathrm{A}-\mathrm{Za}-\mathrm{z}] \\
&\textbf{digit} && \rightarrow [0-9] \\
&\textbf{id} && \rightarrow \textbf{letter}\ (\ \textbf{letter}\ \mid\ \textbf{digit}\ )^*
\end{aligned}$$

Which one of the following Non-deterministic Finite-state Automata with ϵ-transitions accepts the set of valid identifiers? (A double-circle denotes a final state)

A. 

B. 

C. 

D. 

gatecse-2023 theory-of-computation regular-expression one-mark

Answer key☟

6.18.23 Regular Expression: GATE CSE 2024 | Set 1 | Question: 51

Consider the following two regular expressions over the alphabet $\{0, 1\}$:

$r = 0^* + 1^*$

$s = 01^* + 10^*$

The total number of strings of length less than or equal to 5, which are neither in r nor in s, is ________.

gatecse-2024-set1 numerical-answers theory-of-computation regular-expression two-marks

Answer key☟

6.18.24 Regular Expression: GATE CSE 2024 | Set 2 | Question: 52

Let L_1 be the language represented by the regular expression $b^*ab^*(ab^*ab^*)^*$ and $L_2 = \{w \in (a+b)^* || w \mid \leq 4\}$, where $|w|$ denotes the length of string w. The number of strings in L_2 which are also in L_1 is _________.

gatecse-2024-set2 numerical-answers theory-of-computation regular-expression two-marks

Answer key☟

6.18.25 Regular Expression: GATE IT 2004 | Question: 7

Which one of the following regular expressions is NOT equivalent to the regular expression $(a+b+c)^*$?

A. $(a^* + b^* + c^*)^*$
B. $(a^*b^*c^*)^*$
C. $((ab)^* + c^*)^*$
D. $(a^*b^* + c^*)^*$

gateit-2004 theory-of-computation regular-expression normal

Answer key☟

6.18.26 Regular Expression: GATE IT 2005 | Question: 5

Which of the following statements is TRUE about the regular expression 01^*0?

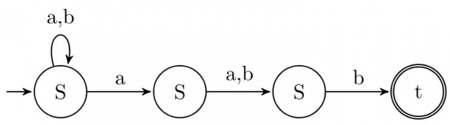
A. It represents a finite set of finite strings.
B. It represents an infinite set of finite strings.
C. It represents a finite set of infinite strings.
D. It represents an infinite set of infinite strings.

gateit-2005 theory-of-computation regular-expression easy

Answer key☟

6.18.27 Regular Expression: GATE IT 2006 | Question: 5

Which regular expression best describes the language accepted by the non-deterministic automaton below?



A. $(a+b)^*\ a(a+b)b$
B. $(abb)^*$
C. $(a+b)^*\ a(a+b)^*\ b(a+b)^*$
D. $(a+b)^*$

gateit-2006 theory-of-computation regular-expression normal

Answer key☟

6.18.28 Regular Expression: GATE IT 2007 | Question: 73

Consider the regular expression $R = (a+b)^*\ (aa+bb)\ (a+b)^*$

Which one of the regular expressions given below defines the same language as defined by the regular expression R ?

A. $(a(ba)^* + b(ab)^*)(a+b)^+$
B. $(a(ba)^* + b(ab)^*)^*(a+b)^*$
C. $(a(ba)^*(a+bb) + b(ab)^*(b+aa))(a+b)^*$
D. $(a(ba)^*(a+bb) + b(ab)^*(b+aa))(a+b)^+$

gateit-2007 theory-of-computation regular-expression normal

Answer key☟

6.18.29 Regular Expression: GATE IT 2008 | Question: 5

Which of the following regular expressions describes the language over $\{0,1\}$ consisting of strings that contain exactly two 1's?

A. $(0+1)^*\ 11(0+1)^*$
B. $0^*\ 110^*$
C. $0^*10^*10^*$
D. $(0+1)^*1(0+1)^*1(0+1)^*$

gateit-2008 theory-of-computation regular-expression easy

Answer key☟

6.19 Regular Grammar (3)

6.19.1 Regular Grammar: GATE CSE 1990 | Question: 15a

Is the language generated by the grammar G regular? If so, give a regular expression for it, else prove otherwise

- G:
  - $S \rightarrow aB$
  - $B \rightarrow bC$
  - $C \rightarrow xB$
  - $C \rightarrow c$

gate1990 descriptive theory-of-computation regular-language regular-grammar

Answer key☟

6.19.2 Regular Grammar: GATE CSE 2015 | Set 2 | Question: 35

Consider the alphabet $\Sigma = \{0,1\}$, the null/empty string λ and the set of strings $X_0, X_1,$ and X_2 generated by the corresponding non-terminals of a regular grammar. $X_0, X_1,$ and X_2 are related as follows.

- $X_0 = 1X_1$
- $X_1 = 0X_1 + 1X_2$
- $X_2 = 0X_1 + \{\lambda\}$

Which one of the following choices precisely represents the strings in X_0?

A. $10(0^* + (10)^*)1$
B. $10(0^* + (10)^*)^*1$
C. $1(0 + 10)^*1$
D. $10(0 + 10)^*1 + 110(0 + 10)^*1$

gatecse-2015-set2 theory-of-computation regular-grammar normal

Answer key

6.19.3 Regular Grammar: GATE IT 2006 | Question: 29

Consider the regular grammar below

$S \rightarrow bS \mid aA \mid \epsilon$
$A \rightarrow aS \mid bA$

The Myhill-Nerode equivalence classes for the language generated by the grammar are

A. $\{w \in (a+b)^* \mid \#a(w) \text{ is even}) \text{ and} \{w \in (a+b)^* \mid \#a(w) \text{ is odd}\}$
B. $\{w \in (a+b)^* \mid \#a(w) \text{ is even}) \text{ and} \{w \in (a+b)^* \mid \#b(w) \text{ is odd}\}$
C. $\{w \in (a+b)^* \mid \#a(w) = \#b(w) \text{and } \{w \in (a+b)^* \mid \#a(w) \neq \#b(w)\}$
D. $\{\epsilon\}, \{wa \mid w \in (a+b)^* \text{and} \{wb \mid w \in (a+b)^*\}$

gateit-2006 theory-of-computation normal regular-grammar

Answer key

6.20 Regular Language (35)

✍ **Practice Tests:** Test 1 (15Q) Test 2 (15Q) Test 3 (15Q) Test 4 (7Q)

6.20.1 Regular Language: GATE CSE 1987 | Question: 2h

State whether the following statements are TRUE or FALSE:

Regularity is preserved under the operation of string reversal.

gate1987 theory-of-computation regular-language true-false

Answer key

6.20.2 Regular Language: GATE CSE 1987 | Question: 2i

State whether the following statements are TRUE or FALSE:

All subsets of regular sets are regular.

gate1987 theory-of-computation regular-language true-false

Answer key

6.20.3 Regular Language: GATE CSE 1990 | Question: 3-viii

Let R_1 and R_2 be regular sets defined over the alphabet Σ Then:

A. $R_1 \cap R_2$ is not regular.
B. $R_1 \cup R_2$ is regular.
C. $\Sigma^* - R_1$ is regular.
D. R_1^* is not regular.

gate1990 normal theory-of-computation regular-language multiple-selects

Answer key

6.20.4 Regular Language: GATE CSE 1991 | Question: 03,xiv

Which of the following is the strongest correct statement about a finite language over some finite alphabet Σ?

A. It could be undecidable
B. It is Turing-machine recognizable
C. It is a context-sensitive language.
D. It is a regular language.
E. None of the above,

gate1991 theory-of-computation easy regular-language multiple-selects

Answer key

6.20.5 Regular Language: GATE CSE 1995 | Question: 2.24

Let $\Sigma = \{0, 1\}$, $L = \Sigma^*$ and $R = \{0^n 1^n \mid n > 0\}$ then the languages $L \cup R$ and R are respectively

A. regular, regular
B. not regular, regular
C. regular, not regular
D. not regular, not regular

gate1995 theory-of-computation easy regular-language

Answer key

6.20.6 Regular Language: GATE CSE 1996 | Question: 1.10

Let $L \subseteq \Sigma^*$ where $\Sigma = \{a, b\}$. Which of the following is true?

a. $L = \{x \mid x \text{ has an equal number of } a\text{'s and } b\text{'s}\}$ is regular
b. $L = \{a^n b^n \mid n \geq 1\}$ is regular
c. $L = \{x \mid x \text{ has more number of } a\text{'s than } b\text{'s}\}$ is regular
d. $L = \{a^m b^n \mid m \geq 1, n \geq 1\}$ is regular

gate1996 theory-of-computation normal regular-language

Answer key

6.20.7 Regular Language: GATE CSE 1998 | Question: 2.6

Which of the following statements is false?

a. Every finite subset of a non-regular set is regular
b. Every subset of a regular set is regular
c. Every finite subset of a regular set is regular
d. The intersection of two regular sets is regular

gate1998 theory-of-computation easy regular-language

Answer key

6.20.8 Regular Language: GATE CSE 1999 | Question: 6

A. Given that A is regular and $(A \cup B)$ is regular, does it follow that B is necessarily regular? Justify your answer.
B. Given two finite automata $M1, M2$, outline an algorithm to decide if $L(M1) \subset L(M2)$. (note: strict subset)

gate1999 theory-of-computation normal regular-language descriptive

Answer key

6.20.9 Regular Language: GATE CSE 2000 | Question: 2.8

What can be said about a regular language L over $\{a\}$ whose minimal finite state automaton has two states?

A. L must be $\{a^n \mid n \text{ is odd}\}$
B. L must be $\{a^n \mid n \text{ is even}\}$
C. L must be $\{a^n \mid n \geq 0\}$
D. Either L must be $\{a^n \mid n \text{ is odd}\}$, or L must be $\{a^n \mid n \text{ is even}\}$

gatecse-2000 theory-of-computation easy regular-language

Answer key

6.20.10 Regular Language: GATE CSE 2001 | Question: 1.4

Consider the following two statements:

$S_1 : \{0^{2n} \mid n \geq 1\}$ is a regular language

$S_2 : \{0^m 1^n 0^{m+n} \mid m \geq 1 \text{ and } n \geq 1\}$ is a regular language

Which of the following statement is correct?

A. Only S_1 is correct
B. Only S_2 is correct
C. Both S_1 and S_2 are correct
D. None of S_1 and S_2 is correct

gatecse-2001 theory-of-computation easy regular-language

Answer key

6.20.11 Regular Language: GATE CSE 2001 | Question: 2.6

Consider the following languages:

- $L1 = \{ww \mid w \in \{a, b\}^*\}$
- $L2 = \{ww^R \mid w \in \{a, b\}^*, w^R \text{ is the reverse of w}\}$
- $L3 = \{0^{2i} \mid \text{ i is an integer}\}$
- $L4 = \{0^{i^2} \mid \text{ i is an integer}\}$

Which of the languages are regular?

A. Only $L1$ and $L2$
B. Only $L2, L3$ and $L4$
C. Only $L3$ and $L4$
D. Only $L3$

gatecse-2001 theory-of-computation normal regular-language

Answer key

6.20.12 Regular Language: GATE CSE 2007 | Question: 31

Which of the following languages is regular?

A. $\{ww^R \mid w \in \{0, 1\}^+\}$
B. $\{ww^R x \mid x, w \in \{0, 1\}^+\}$
C. $\{wxw^R \mid x, w \in \{0, 1\}^+\}$
D. $\{xww^R \mid x, w \in \{0, 1\}^+\}$

gatecse-2007 theory-of-computation normal regular-language

Answer key

6.20.13 Regular Language: GATE CSE 2007 | Question: 7

Which of the following is TRUE?

A. Every subset of a regular set is regular
B. Every finite subset of a non-regular set is regular
C. The union of two non-regular sets is not regular
D. Infinite union of finite sets is regular

gatecse-2007 theory-of-computation easy regular-language

Answer key

6.20.14 Regular Language: GATE CSE 2008 | Question: 53

Which of the following are regular sets?

I. $\{a^n b^{2m} \mid n \geq 0, m \geq 0\}$
II. $\{a^n b^m \mid n = 2m\}$
III. $\{a^n b^m \mid n \neq m\}$

IV. $\{xcy \mid x, y, \in \{a, b\}^*\}$

A. I and IV only
B. I and III only
C. I only
D. IV only

gatecse-2008 theory-of-computation normal regular-language

Answer key☟

6.20.15 Regular Language: GATE CSE 2011 | Question: 24

Let P be a regular language and Q be a context-free language such that $Q \subseteq P$. (For example, let P be the language represented by the regular expression p^*q^* and Q be $\{p^nq^n \mid n \in N\}$). Then which of the following is **ALWAYS** regular?

A. $P \cap Q$
B. $P - Q$
C. $\Sigma^* - P$
D. $\Sigma^* - Q$

gatecse-2011 theory-of-computation easy regular-language

Answer key☟

6.20.16 Regular Language: GATE CSE 2012 | Question: 25

Given the language $L = \{ab, aa, baa\}$, which of the following strings are in L^*?

1. $abaabaaabaa$
2. $aaaabaaaa$
3. $baaaaabaaaab$
4. $baaaaabaa$

A. $1, 2 \text{ and } 3$
B. $2, 3 \text{ and } 4$
C. $1, 2 \text{ and } 4$
D. $1, 3 \text{ and } 4$

gatecse-2012 theory-of-computation easy regular-language

Answer key☟

6.20.17 Regular Language: GATE CSE 2013 | Question: 8

Consider the languages $L_1 = \phi$ and $L_2 = \{a\}$. Which one of the following represents $L_1{L_2}^* \cup {L_1}^*$?

A. $\{\epsilon\}$
B. ϕ
C. a^*
D. $\{\epsilon, a\}$

gatecse-2013 theory-of-computation normal regular-language

Answer key☟

6.20.18 Regular Language: GATE CSE 2014 | Set 1 | Question: 15

Which one of the following is **TRUE**?

A. The language $L = \{a^nb^n \mid n \geq 0\}$ is regular.
B. The language $L = \{a^n \mid n \text{ is prime }\}$ is regular.
C. The language L= $\{w \mid w \text{ has } 3k + 1 \; b's \text{ for some } k \in N \text{ with } \Sigma = \{a, b\}\}$ is regular.
D. The language $L = \{ww \mid w \in \Sigma^* \text{ with } \Sigma = \{0, 1\}\}$ is regular.

gatecse-2014-set1 theory-of-computation regular-language normal

Answer key☟

6.20.19 Regular Language: GATE CSE 2014 | Set 2 | Question: 15

If $L_1 = \{a^n \mid n \geq 0\}$ and $L_2 = \{b^n \mid n \geq 0\}$, consider

I. $L_1.L_2$ is a regular language
II. $L_1.L_2 = \{a^nb^n \mid n \geq 0\}$

Which one of the following is CORRECT?

A. Only I
B. Only II
C. Both I and II
D. Neither I nor II

gatecse-2014-set2 theory-of-computation normal regular-language

Answer key☟

6.20.20 Regular Language: GATE CSE 2014 | Set 2 | Question: 36

Let $L_1 = \{w \in \{0,1\}^* \mid w$ has at least as many occurrences of $(110)'$s as $(011)'$s$\}$. Let $L_2 = \{w \in \{0,1\}^* \mid w$ has at least as many occurrences of $(000)'$s as $(111)'$s$\}$. Which one of the following is TRUE?

A. L_1 is regular but not L_2
B. L_2 is regular but not L_1
C. Both L_1 and L_2 are regular
D. Neither L_1 nor L_2 are regular

gatecse-2014-set2 theory-of-computation normal regular-language

Answer key☟

6.20.21 Regular Language: GATE CSE 2015 | Set 2 | Question: 51

Which of the following is/are regular languages?

$L_1 : \left\{wxw^R \mid w, x \in \{a,b\}^* \text{ and } |w|, |x| > 0\right\}, w^R \text{ is the reverse of string } w$

$L_2 : \{a^n b^m \mid m \neq n \text{ and } m, n \geq 0\}$

$L_3 : \{a^p b^q c^r \mid p, q, r \geq 0\}$

A. L_1 and L_3 only
B. L_2 only
C. L_2 and L_3 only
D. L_3 only

gatecse-2015-set2 theory-of-computation normal regular-language

Answer key☟

6.20.22 Regular Language: GATE CSE 2016 | Set 2 | Question: 17

Language L_1 is defined by the grammar: $S_1 \rightarrow aS_1b \mid \varepsilon$

Language L_2 is defined by the grammar: $S_2 \rightarrow abS_2 \mid \varepsilon$

Consider the following statements:

- P: L_1 is regular
- Q: L_2 is regular

Which one of the following is **TRUE**?

A. Both P and Q are true.
B. P is true and Q is false.
C. P is false and Q is true.
D. Both P and Q are false.

gatecse-2016-set2 theory-of-computation easy regular-language

Answer key☟

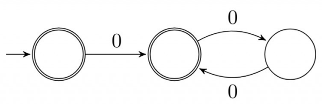
6.20.23 Regular Language: GATE CSE 2018 | Question: 52

Given a language L, define L^i as follows:

$$L^0 = \{\varepsilon\}$$

$$L^i = L^{i-1} \bullet L \text{ for all } I > 0$$

The order of a language L is defined as the smallest k such that $L^k = L^{k+1}$. Consider the language L_1(over alphabet 0) accepted by the following automaton.



The order of L_1 is ________.

gatecse-2018 theory-of-computation numerical-answers regular-language two-marks

Answer key

6.20.24 Regular Language: GATE CSE 2019 | Question: 7

If L is a regular language over $\Sigma = \{a, b\}$, which one of the following languages is NOT regular?

A. $L.L^R = \{xy \mid x \in L, y^R \in L\}$
B. $\{ww^R \mid w \in L\}$
C. $\text{Prefix}\ (L) = \{x \in \Sigma^* \mid \exists y \in \Sigma^* \text{such that } xy \in L\}$
D. $\text{Suffix}\ (L) = \{y \in \Sigma^* \mid \exists x \in \Sigma^* \text{such that } xy \in L\}$

gatecse-2019 theory-of-computation regular-language one-mark

Answer key

6.20.25 Regular Language: GATE CSE 2020 | Question: 51

Consider the following language.

$L = \{x \in \{a, b\}^* \mid \text{number of } a\text{'s in } x \text{ divisible by } 2 \text{ but not divisible by } 3\}$

The minimum number of states in DFA that accepts L is _________

gatecse-2020 numerical-answers theory-of-computation regular-language two-marks

Answer key

6.20.26 Regular Language: GATE CSE 2020 | Question: 8

Consider the following statements.

I. If $L_1 \cup L_2$ is regular, then both L_1 and L_2 must be regular.
II. The class of regular languages is closed under infinite union.

Which of the above statements is/are TRUE?

A. I only
B. II only
C. Both I and II
D. Neither I nor II

gatecse-2020 theory-of-computation regular-language one-mark

Answer key

6.20.27 Regular Language: GATE CSE 2021 | Set 2 | Question: 9

Let $L \subseteq \{0, 1\}^*$ be an arbitrary regular language accepted by a minimal DFA with k states. Which one of the following languages must necessarily be accepted by a minimal DFA with k states?

A. $L - \{01\}$
B. $L \cup \{01\}$
C. $\{0,1\}^* - L$
D. $L \cdot L$

gatecse-2021-set2 theory-of-computation finite-automata regular-language one-mark

Answer key

6.20.28 Regular Language: GATE CSE 2024 | Set 1 | Question: 13

Let L_1, L_2 be two regular languages and L_3 a language which is not regular.
Which of the following statements is/are always TRUE?

A. $L_1 = L_2$ if and only if $L_1 \cap \overline{L_2} = \phi$
B. $L_1 \cup L_3$ is not regular
C. $\overline{L_3}$ is not regular
D. $\overline{L_1} \cup \overline{L_2}$ is regular

gatecse-2024-set1 multiple-selects theory-of-computation regular-language one-mark

Answer key

6.20.29 Regular Language: GATE CSE 2025 | Set 1 | Question: 18

A regular language L is accepted by a non-deterministic finite automaton (NFA) with n states. Which of the following statement(s) is/are FALSE?

A. L may have an accepting NFA with $< n$ states.
B. L may have an accepting DFA with $< n$ states.
C. There exists a DFA with $\leq 2^n$ states that accepts L.
D. Every DFA that accepts L has $> 2^n$ states.

gatecse2025-set1 theory-of-computation finite-automata regular-language multiple-selects one-mark

Answer key

6.20.30 Regular Language: GATE CSE 2025 | Set 1 | Question: 34

Consider the following two languages over the alphabet $\{a, b\}$:

$$L_1 = \{\alpha\beta\alpha \mid \alpha \in \{a, b\}^+ \text{AND } \beta \in \{a, b\}^+\}$$
$$L_2 = \{\alpha\beta\alpha \mid \alpha \in \{a\}^+ \text{AND } \beta \in \{a, b\}^+\}$$

Which ONE of the following statements is CORRECT?

A. Both L_1 and L_2 are regular languages.
B. L_1 is a regular language but L_2 is not a regular language.
C. L_1 is not a regular language but L_2 is a regular language.
D. Neither L_1 nor L_2 is a regular language.

gatecse2025-set1 theory-of-computation regular-language two-marks

Answer key

6.20.31 Regular Language: GATE CSE 2025 | Set 2 | Question: 42

Let $\Sigma = \{a, b, c\}$. For $x \in \Sigma^*$, and $\alpha \in \Sigma$, let $\#_\alpha(x)$ denote the number of occurrences of α in x.
Which one or more of the following option(s) define(s) regular language(s)?

A. $\{a^m b^n \mid m, n \geq 0\}$
B. $\{a, b\}^* \cap \{a^m b^n c^{m-n} \mid m \geq n \geq 0\}$
C. $\{w \mid w \in \{a, b\}^*, \#_a(w) \equiv 2 (\bmod 7)$, and $\#_b(w) \equiv 3 (\bmod 9)\}$
D. $\{w \mid w \in \{a, b\}^*, \#_a(w) \equiv 2 (\bmod 7)$, and $\#_a(w) = \#_b(w)\}$

gatecse2025-set2 theory-of-computation regular-language multiple-selects two-marks

Answer key

6.20.32 Regular Language: GATE IT 2006 | Question: 30

Which of the following statements about regular languages is NOT true ?

A. Every language has a regular superset
B. Every language has a regular subset
C. Every subset of a regular language is regular
D. Every subset of a finite language is regular

gateit-2006 theory-of-computation easy regular-language

Answer key